Department of Chemistry, IIT Jodhpur

BS in Chemistry with Specialization in a Focused Area

1. Introduction

Vision of the Department of Chemistry is to achieve academic excellence and be acknowledged for high quality fundamental, sustainable, and translational research in chemical sciences and allied research areas that addresses continuously evolving current global issues on energy, food, water, and healthcare. Along this line, the department has initiated an engineering-linked four-year undergraduate Bachelor of Science (BS) in chemistry program with specialization in a focused area to attract and foster young and aspiring minds who are driven by curiosity and interested in cutting-edge research and technologies across the disciplines of chemical science. The central theme of this program is science-based engineering innovation and through this undergraduate programme in chemistry at IIT Jodhpur, we are striving to cross the barrier of traditional paths and planning to establish ourselves to become a leading department at the interplay of science and technology. This uniquely blended program is along the line of National Education Policy, as well as the Institute's vision of holistic development of students.

This undergraduate program is designed to transmit fundamental knowledge in basic science and engineering during the first few semesters, which develops critical analytical thinking and forms the basis of diverse skill sets used by scientists and engineers to innovate, design, and develop technologies for the future. Students undergo comprehensive training in advanced concepts of chemistry for the next two years. The inclusion of a substantial component of courses from other science and engineering departments makes the program quite interdisciplinary in nature. The final year of the curriculum is designed towards specialization in a focused area. The flexibility of opting for different specializations is ingrained in formulating the structure and curriculum of the programme. During this year, the students get to experience the joys and rigors of original research currently being pursued and consider a variety of career paths through capability-linked specializations of their choice. In the specialization, emphasis is given to the emerging areas in Chemistry to meet societal and industrial needs, and ever-growing sustainability demands. The graduating students of this program shall be equipped with requisite skill-sets required to assume leadership positions in the different sectors of industry, further their career in higher studies, or embark on entrepreneurial ventures to address present day unmet challenges of the country.

2. Objectives of the program

The objectives of this program are:

- to equip new generation aspiring minds with engineering linked chemical science skill sets enabling them to opt for challenging career paths at various levels of academia and industry;

- to impart education towards understanding of the concepts of chemical science, basic engineering, and allied areas enabling the students to tackle societal problems through an interdisciplinary approach;

- to integrate analytical and computational abilities with experimental skills to create individuals competent in professional science practices;

- to facilitate and promote entry into research at an early stage through a flexible and borderless curriculum;

- to inculcate an attitude towards commitment to scientific ethics, leadership qualities, entrepreneurship, and professional development for the future generation;

- to instil the approach of collaborative research between academia and industry targeting sustainability.

3. Expected Graduate Attributes:

Graduates of the program are expected to have following attributes:

- Strong understanding of fundamentals of science and engineering with emphasis on chemical science.

- A strong foundation in the analytical, experimental, and computational methods to study various aspects of chemical science.

- Foundations for critical thinking that are needed to broaden their careers in diverse cross-disciplines.

- Ability to acquire new knowledge, integrate in the existing paradigm, and innovate solutions as per the requirement.

- Capability to appreciate synthetic methods for the development of important chemical products such as pharmaceuticals, agrochemicals, paints, petrochemicals, food additives, plastics, etc.

- Ability to handle analytical, spectroscopic, microscopic and other instrumentation methods for interdisciplinary research areas.

- Use of computational tools to understand and analyse chemical and biophysical processes, molecular structures and advanced materials.

- Ability to comprehend, write, and present scientific articles and technical reports.

- Ability to engage in ethical and responsible practices while pursuing professional growth.

4. Learning Outcomes

- Acquire strong fundamentals in theoretical and practical aspects of organic, inorganic and physical chemistry as well as other branches of basic science and engineering.
- Apply the concepts of science and basic engineering to analyse and design complex novel materials, and molecular complexes having potential applications to society.
- Develop skills on hands-on wet chemistry, analytical, instrumentation, and experimental data handling to tackle scientific problems.

- Perform literature review and patent landscaping for innovative research.

- Develop communication skills by participating in classroom presentations, seminars and workshops on manuscript / patent writing / presentation skills.

- Value of team work and group discussion in problem solving, and time management to continue co-curricular activities without compromising academics.

- To understand professional ethics and social responsibilities.

5. New skill sets targeted

Industry-R&D-Academia collaboration on cutting edge science and technologies is the key for the growth of the country. Hence students need to delve into domains ranging from sustainable materials synthesis to molecular complex design, organic synthesis for pharmaceuticals, agriculture industries to industrial Catalysis that are critical to meet future needs. In the pursuit of novel materials, molecules and catalysts relevant for today's Circular Chemistry (Recycle and Reuse, rather than take-make-dispose) that enables a circular economy requires comprehensive understanding of the following specialized areas:

- Spectroscopy and Dynamics
- Organic and Organometallic Catalysis
- Advanced Energy Materials
- Entrepreneurship

<u>Proposed Specialization Areas</u>

1. Spectroscopy and Dynamics
2. Organic and Organometallic Catalysis
3. Advanced Energy Materials
4. Entrepreneurship

Program Structure

S. No.	Course Type	Course Category	Credits	
1.	Institute Core (I)	Engineering (IE)	29	**70**
		Science (IS)	18	
		Humanities (IH)	12	
2.	Programme Linked (L)	Programme linked courses (PL)	11	
3.	Programme Core (P)	Programme Compulsory (PC) + BS Project (PP)	60	**60**
4.	Specialization (E)	Specialization Core (SC)	12	**30**
		Specialization Elective (SE)	18	
5.	Non-Graded (N)	Humanities (NH)	6	**15**
		Engineering (NE)	3	
		Design/Practical Experience (ND)	6	
			Total	**175**

- Those who are opting for specialization in Entrepreneurship will be required to earn 10 credits via relevant specialization electives.

Programme Core (PC) Courses

S. No.	Course Name & Course Code	LTP	Contact Hours	Credit
1.	Basic Organic Chemistry	3-0-0	3	3
2.	Physical Organic Chemistry and Photochemistry	3-0-0	3	3
3.	Purification and Characterization of Organic Molecules	3-0-0	3	3
4.	Biochemistry	3-0-0	3	3
5.	Fundamental Inorganic Chemistry	3-0-0	3	3
6.	Chemistry of Elements	3-0-0	3	3
7.	Coordination and Bioinorganic Chemistry	3-0-0	3	3
8.	Organometallics	3-0-0	3	3
9.	Chemical Kinetics, Photochemistry and Theory of Solutions	3-0-0	3	3
10.	Structural Chemistry	3-0-0	3	3
11.	Atomic and Molecular Structure	3-0-0	3	3
12.	Fundamentals of Molecular Spectroscopy	3-0-2	5	4
13.	Thermodynamics, Chemical Equilibria, Phase Equilibria and Electrochemistry	3-0-0	3	3
14.	Statistical Mechanics	2-0-0	2	2
15.	Entrepreneurship for Chemists	2-0-0	2	2
16.	BS Project	0-0-3	3	3
17.	Organic Chemistry Lab 1	0-0-3	3	1.5
18.	Organic Chemistry Lab 2	0-0-3	3	1.5
19.	Inorganic Chemistry Lab 1	0-0-3	3	1.5
20.	Inorganic Chemistry Lab 2	0-0-3	3	1.5
21.	Physical Chemistry Lab 1	0-0-3	3	1.5
22.	Physical Chemistry Lab 2	0-0-3	3	1.5
23.	Basic Computational Lab	0-0-3	3	1.5
24.	Applied Computational Lab	0-0-3	3	1.5
25.	Materials Characterization Lab	0-0-2	2	1
			Total	**60**

Topic Clouds and Mapping of Topic Clouds with Proposed Courses

Area	Category	Topics	Course (IE/PC/PE)
Organic Chemistry	Core	Hybridization, Nomenclature, Functional groups, intermediates, transition states, organic reactions, reaction mechanism, acids and bases, conformational analysis and stereochemistry, stereo electronic effects, pericyclic reactions, photochemistry, spectroscopy in organic structure elucidation, biomolecules, carbohydrate, lipids, amino acids, proteins, peptides	Basic Organic Chemistry; Physical Organic Chemistry and Photochemistry; Biochemistry
	Techniques	IR, UV-Vis, MP, 1H-NMR, 13C-NMR, Thin layer chromatography, Column Chromatography, HPLC, High Resolution Mass Spectrometry, Gas Chromatography	Purification and Characterization of organic molecules
	Applications	Identifications, Structural elucidation of organic compounds, catalysis and methodology development, synthesis of natural and unnatural compounds	Purification and Characterization of organic molecules; Physical Organic Chemistry and Photochemistry
Inorganic Chemistry	Core	Transition Metal, Coordination chemistry, and Organometallics, Solid -State and Materials Chemistry, Main Group Chemistry and Bioinorganic Chemistry, Environmental Chemistry, Catalysis	Fundamental Inorganic Chemistry, Chemistry of elements, Coordination and Bioinorganic Chemistry
	Techniques	SCXRD, PXRD, NMR, IR, TGA, BET, XPS UV-Vis, RAMAN, CV, Thin film Synthesis, Deposition and growth methods, Lithography, Mass Spectroscopy, Atomic Absorption Spectroscopy, ICP-OES	Chemistry of Elements, Characterization techniques in Inorganic Chemistry
	Applications	Catalysis, Functional Electronic Materials Magnetic Materials, Thermoelectric, Energy Storage, Solar Cells, H_2 economy, sustainable energy, Environment and Technology	Chemistry of elements, Coordination and Bioinorganic Chemistry

Physical Chemistry	Core	Atomic structure; Solutions, Surface Chemistry, Electrochemistry, Thermodynamics, Chemical Kinetics, Chemical Equilibrium, Laws of Gasses, Spectroscopy, Quantum Chemistry, Gaseous state, liquid state, solid state Phase equilibrium, Degree of freedom, Gibbs Phase Rule, Electrode potential, Galvanic cell, Battery, Surface chemistry.	Statistical Mechanics; Atomic and Molecular Structure
	Techniques	DLS, Zeta Potential, UV-Vis spectroscopy, Fluorescence, FTIR, NMR, Electroanalytical Techniques, XPS, Numerical techniques, Microscopy	Fundamentals of Molecular Spectroscopy; Mathematics I
	Applications	Exactly solvable problems, Different state of matter with industrial applications, Energy Storage, Solar Cells, Fuel Cells, Sensors,	Atomic and Molecular Structure; Chemistry of Elements

Specializations

S. No.	Specialization	Description	
1.	Spectroscopy and Dynamics	Acquiring a detailed understanding of interactions between molecules, mechanism and dynamics of chemical reactions, and many body natural phenomena are of great importance to Chemistry. The specialization on Spectroscopy and Dynamics aims to build on a fundamental understanding of wide-ranging aspects of the area covering Magnetic Resonance (MR), Fluorescence and single molecule spectroscopy, classical and quantum chemical calculations and quantum computation to study dynamics of species ranging from diatomics, ionic species, macromolecules, and complex fluids via large scale problem solving. Specialization in this area is expected to enable graduates to explain chemical and physical phenomena at the molecular level and prepare graduates with hands-on experience on spectroscopic techniques and computational methods.	
		Specialization Core (12 credits)	**Specialization Elective (18 credits)**
		Numerical Methods and Group Theory (CYL6330) Theory of Complex Fluids Chemical Bonding and Statistical Rate Theories Theoretical Account of Spectroscopy	Introduction to Soft Matter Physics (PHL6330) Modern NMR Methods for Chemist (CYL6370) Nuclear Magnetic Resonance: Principles and Practices (CYL7090) Principles of Fluorescence Spectroscopy and Imaging (CYL6360) Single Molecule Spectroscopy Molecular Dynamics Simulations (CYL7310) Modern Electronic Structure Theory (CYL7380) Chemical Reaction Dynamics (CYL7390) Stochastic Problems in Biophysics (CYL7410) Polymer Dynamics (CYL7400)

			Statistical Mechanics and Molecular Simulations (CYL7500) Computational Lab (CYP6060) Modern Physics Lab Theory and Applications of Spectroscopic Techniques (CYL6090) Chemical Binding (CYL6320) Project (6 credits) Advanced Fluorescence Microscopy Techniques and Applications
Area	**Category**	**Topics**	**Course**
Spectroscopy and Dynamics	Core	Probability theory and statistics, Born-Oppenheimer Approximation; Electronic Schrodinger equation; Approximate methods;	Chemical Bonding and Statistical Rate Theories; Theory of Complex Fluids, Stochastic Problems in Biophysics, Polymer Dynamics
	Techniques	Hartree-Fock method; Slater determinants; Roothan solution; Molecular simulations,	Chemical Bonding and Statistical Rate Theories, Molecular Dynamics Simulations; Statistical Mechanics and Molecular Simulations
	Applications	Electronic structure; Finding equilibrium geometries and energies; Transition state finding;	Chemical Bonding and Statistical Rate Theories; Computational Lab; Introduction of soft matter Physics; Project
		Catalysis in general, offers a unique synthetic tool to synthesize and manufacture a diverse variety of synthetic compounds in an environment-friendly and sustainable fashion. Organic and organometallic catalysis, a major pillar of catalysis, is of utmost importance in different industries, such as pharmaceutical, agrochemical, petrochemical, biomaterial and nanotechnology. This specialization is aimed to educate students regarding different facets of organic and organometallic catalysis and their importance in an array of efficient and selective synthetic transformations leading to the synthesis of	

2.	Organic and Organometallic Catalysis	value-added chemicals, biologically active compounds, and natural products.	
		Specialization Core (12 credits)	**Specialization Elective (18 credits)**
		Application of Organometallic Compounds in Synthesis Organic Synthesis (CYL6060) Catalysis in Organic Synthesis Industrial Catalysis	Project (6 credits) Characterization Techniques in Inorganic Chemistry (CYL7420) Compounds of P-Block Elements and Applications (CYL7470) Frontiers in Organometallic Catalysis (CYL7370) Frontiers in Organic Chemistry (CYL7340) Isolation and Synthesis of Natural Products (CYL7460) Stereochemistry of Organic Compounds (CYL7440) Heterocyclic Chemistry (CYL7XXX) Catalysis for Energy (CYL7330) Advanced Catalysis (CYL7XXX) Advanced Main Group Chemistry (CYL6350) Molecular Modelling for Catalytic Reactions Sustainable Catalysis
Area	**Category**	**Topics**	**Course**
	Core	Organic and organometallic compounds, reagents, catalysts,	Organic Synthesis; Catalysis in Organic Synthesis; Application of Organometallic compounds in synthesis, Industrial Catalysis
	Techniques	Characterization, NMR, IR, Mass, UV, EPR, CV, Raman, Mossbauer, XRD, theoretical calculations	Characterization Techniques in Inorganic Chemistry

Organic and Organometallic Catalysis			
	Applications	industrially relevant bulk and fine chemicals, natural products, pharmaceutical, agrochemicals, sustainable transformations, alternative energy, Hydrogen storage, asymmetric synthesis	Frontiers in Organometallic Catalysis; Frontiers in Organic Chemistry; Isolation and Synthesis of Natural Products; Advanced Catalysis; Industrial Catalysis; Heterocycle Chemistry; Project
3.	Advanced Energy Materials	Energy materials encompasses all the aspects of energy harvesting materials and their corresponding devices. Here, various renewable and non-renewable energy sources and their corresponding devices will be studied. The principle behind their operation and how the efficiency of the devices can be optimized to achieve maximum output from the devices would be discussed. This specialization will provide an in-depth understanding of the energy materials and their future applications.	
		Specialization Core (12 credits)	**Specialization Elective (18 credits)**
		1. Materials for Energy Technologies 2. Advanced Electrochemistry and Applications 3. Fabrication and Characterization of Devices 4. Electrical Energy Storage Materials and Devices 5. Nanomaterials and Nanodevices	1. Physics of Solar Cells 2. Optics for Solar Energy Applications 3. Nanomaterials for Hydrogen Production and Storage 4. Bioenergy 5. Energy Harvesting 6. Nanoscience and Nanotechnology 7. Hydrogen Generation, Storage and Applications 8. Materials Management for Energy Technology 9. Solar Resource Assessment and Its Management 10. Project (6 credits)
Area	**Category**	**Topics**	**Course**
Advanced Energy Materials	Core	Conventional energy systems and related materials challenges, transformer materials, Solar thermal energy systems, materials for concentrator, reflectors, heat transfer and storage, solar photovoltaics, materials for energy conversion, materials for electrical energy storage, wind energy, materials for wind blades, nuclear energy, geothermal energy	Materials for Energy Technologies

		Historical background, rechargeable and non-rechargeable batteries, primary and secondary battery, lead-acid battery, nickel-cadmium battery, metal ion batteries (Lithium, sodium and potassium), multivalent metal ion batteries (magnesium, aluminum, zinc), metal-air battery, redox-flow battery, photo rechargeable batteries, Technical challenges, liquid, gel, solid and solid polymer electrolytes, supercapacitors, modeling electrode materials and the physics of electrolytes.	Electrical Energy Storage Materials and Devices
	Techniques	Basics of nanomaterials, nanoparticles, and clusters, quantum confinement and quantum dots, nanotechnology for energy generation and storage: DSSC and QDSSC, fuel cells, hydrogen storage in carbon nanomaterials, nanocatalyst	Nanoscience and Nanotechnology PHL7441 and PHL7442
		Quantum efficiency study of photovoltaics, electrochemical characterizations of energy storage devices, electrical characterization of thermoelectric, piezoelectric, and triboelectric devices. Characterization of PV modules, characterization of battery packs/modules.	Characterization of Energy Devices
		First generation silicon solar cells, thin-film silicon solar cells, CIGS and CdTe solar cells, DSSC, perovskite solar cells, piezoelectric MEMS energy harvester, piezoelectric energy	Energy Harvesting

		transducers, triboelectric nanogenerator, thermoelectric energy harvesting, pyroelectric energy harvesting	
	Applications	Thermodynamics of solar cells and light, properties of semiconductors, generation and recombination of charge carriers, different types of junctions, understanding the current flow across the junction.	Physics of Solar Cells
		Ray optics, design of lenses, concentrating systems, sun simulators, field design for solar thermal application, Hydrogen production from biowaste and fossil fuels, Thermochemical, electrochemical and photo-electrochemical process for hydrogen production, Hydrogen storage in Metal Hydrides, Non-metal hydride nanomaterials for Hydrogen Storage, Thermodynamics of Hydrogen Storage	Optics for Solar Energy Applications
		Introduction, hydrogen production processes, metal hybrid nanomaterials for hydrogen storage, non-metal hydride nanomaterials for hydrogen storage, thermodynamics of hydrogen storage	Nanomaterials for Hydrogen Production and Storage
		Electrochemical dissociation, electrocatalysis, photocatalysis, pressurized hydrogen, liquid hydrogen, metal hydrides, chemisorption materials, fuel cell principles, distributed application for energy generation, electric vehicles	Hydrogen Generation, Storage and Applications
		High-temperature materials for conventional energy, physics of polymer nanocomposite, liquid electrolytes, damage assessment of the materials, recycling of	Materials Management for Energy Technology

		materials used in power plants, case studies,	
		History of bioenergy, bio-energy harvesting, thermoelectrics, bio-reactors, bio-piezoelectric materials, light-harvesting biomaterials, bio-fuel cell, artificial leaf, bio-fluorescent materials, bio-phosphorescent, microbial fuel cells, bio-macromolecules and bio-polymers	Bioenergy
		Solar radiation and measurement, devices for measurement, resource data centers, data assessment, prediction using machine learning and artificial intelligence	Solar Resource Assessment and Its Management

BS in Chemistry with Specialization in a Focused Area – Curriculum

Cat	*Course*	*LTP*	*CH*	*NC*	*GC*	*Cat*	*Course*	*LTP*	*CH*	*NC*	*GC*
	I Semester						**II Semester**				
IE	Introduction to Electrical Engineering	3-0-2	5	-	4	IE	Engineering Mechanics	2-1-0	3	-	3
IE	Introduction to Computer Science	3-0-2	5	-	4	IS	Chemistry	3-0-0	3	-	3
IE	Introduction to Bioengineering	3-0-2	5	-	4	IS	Physics	3-0-0	3	-	3
						IS	Chemistry Lab	0-0-2	2	-	1
						IS	Physics Lab	0-0-2	2	-	1
IS	Mathematics I	3-1-0	4	-	4	IS	Mathematics II	3-1-0	4	-	4
IE	Engineering Visualization	0-0-2	2	-	1	IE	Engineering Realization	0-0-2	2	-	1
NE	Engineering Design I	0-0-2	2	1	-	NE	Engineering Design II	0-0-2	2	1	-
NH	Communication Skill I	0-0-2	2	1	-	NH	Communication Skill II	0-0-2	2	1	-
NH	Social Connect and responsibilities I	0-0-1	1	0.5	-	NH	Social Connect and responsibilities II	0-0-1	1	0.5	-
NH	Performing Arts I /Sports I	0-0-1	1	0.5	-	NH	Performing Arts II/Sports II	0-0-1	1	0.5	-
	Total	**12-1-14**	**27**	**3**	**17**		**Total**	**11-2-12**	**25**	**3**	**16**
	III Semester						**IV Semester**				
PL	Scientific Computation	3-0-2	5	-	4	PC	Thermodynamics, Chemical Equilibria, Phase Equilibria, and Electrochemistry	3-0-0	3	-	3
PC	Basic Organic Chemistry	3-0-0	3	-	3	PC	Chemistry of Elements	3-0-0	3	-	3
IE	Introduction to Machine Learning	3-0-2	5	-	4	PL	Modelling and Simulation	3-0-2	5	-	4
PC	Fundamental Inorganic Chemistry	3-0-0	3	-	3	IE	Data Structure and Algorithms	3-0-2	5	-	4
IE	Signals and Systems	3-1-0	4	-	4	PL	Design and Analysis of Experiments	3-0-0	3	-	3
PC	Organic Chemistry Lab 1	0-0-3	3	-	1.5	PC	Inorganic Chemistry Lab 1	0-0-3	3	-	1.5
PC	Basic Computational Lab	0-0-3	3	-	1.5	PC	Physical Chemistry Lab 1	0-0-3	3	-	1.5
NE	Intro. To Profession	0-0-2	2	1	-	IH	Humanities I	3-0-0	3	-	3

	Total	**15-1-12**	**28**	**1**	**21**		**Total**	**18-0-10**	**28**	**-**	**23**
	V Semester						**VI Semester**				
PC	Biochemistry	3-0-0	3	-	3	PC	Fundamentals of Molecular Spectroscopy	3-0-2	5	-	4
PC	Physical Organic Chemistry and Photochemistry	3-0-0	3	-	3	PC	Organometallics	3-0-0	3	-	3
PC	Atomic and Molecular Structure	3-0-0	3	-	3	PC	Purification and Characterization of Organic Molecules	3-0-0	3	-	3
PC	Statistical Mechanics	2-0-0	2	-	2	PC	Chemical Kinetics, Photochemistry, and Theory of Solutions	3-0-0	3	-	3
PC	Coordination and Bioinorganic Chemistry	3-0-0	3	-	3	PC	BS Project	0-0-3	3	-	3
PC	Applied Computational Lab	0-0-3	3	-	1.5	PC	Entrepreneurship for Chemists	2-0-0	2	-	2
PC	Materials Characterization Lab	0-0-2	2	-	1	PC	Inorganic Chemistry Lab 2	0-0-3	3	-	1.5
PC	Organic Chemistry Lab 2	0-0-3	3	-	1.5	PC	Physical Chemistry Lab 2	0-0-3	3	-	1.5
IH	Humanities II	3-0-0	3	-	3						
NH	Professional Ethics I	0-1-0		1	-	NH	Professional Ethics II	0-0-2	2	1	-
	Total	**17-1-8**	**25**	**1**	**21**		**Total**	**13-0-14**	**27**	**1**	**21**
	VII Semester						**VIII Semester**				
SC	Specialization Core 1	3-0-0	3	-	3	SE	Specialization Elective	-	-	-	18
SC	Specialization Core 2	3-0-0	3	-	3						
SC	Specialization Core 3	3-0-0	3	-	3						
SC	Specialization Core 4	3-0-0	3	-	3						
PC	Structural Chemistry	3-0-0	3	-	3						
IH	Humanities III	3-0-0	3	-	3	IH	Humanities IV	3-0-0	3	-	3
IS	Environmental Science	2-0-0	2	-	2						
	Total	**20-0-0**	**20**	**-**	**20**		**Total**	**-**	**-**	**-**	**21**
										Total of Graded Credit	**-**	**160**
										Total of Non-Graded Credit	**9**	**-**
										Non-Graded Design Credits	**6**	**-**
										Grand Total	**175**	

Detailed Program Compulsory Course Contents

Title	**Basic Organic Chemistry**	Number	CYL2XXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The instructor will:

1. Provide insights into the mechanism of a diverse set of organic reactions representing the structures of organic molecules, its active sites and reactivity.
2. Provide knowledge of fundamental organic chemistry with regard to efficiency and selective of reactions, conformational aspects and stereochemistry.
3. Provide knowledge of properties, preparations, reactions for selective classes of molecules

Learning Outcomes

The students are expected to:

1. Understand the reaction pathways and rationale behind drawing a reaction mechanism.
2. Appreciate the crucial role conformations and configurations in determining the course of organic reactions.
3. Apply the knowledge in predicting reactions and products in organic chemistry.

Contents

Introduction: Hybridization, structures, molecular orbitals, IUPAC nomenclatures of organic molecules **(2 lectures)**.

Mechanism and Intermediates: Reaction mechanisms and different methods for their determination, transition states and intermediates, structure and reactivity of carbocation including nonclassical carbocation, carbanion, radical, carbene, nitrene, concerted and stepwise mechanism, early and late transition states, kinetically controlled and thermodynamically controlled products, exothermic and endothermic reactions with examples **(12 lectures)**.

Conformational Analysis and basic stereochemistry: Conformational analysis of aliphatic acyclic compounds, angle strain, torsional strain, ring strain, conformation of aliphatic cyclic compounds, such as cyclopropane, cyclobutane, cyclopentane and cyclohexane, basics of stereochemistry, enantiomer, diastereomer, epimer, optical activity, enantiomeric excess, D/L notations, R/S notations **(12 lectures)**

Hydrocarbons and Aromatic compounds: Classification and properties of hydrocarbons: alkane, alkene, alkyne, allene; synthesis and their reactions; their industrial source and applications in daily life; aromaticity, Reactions of aromatic compounds, electrophilic substitution, activated nucleophilic substitution, functional group interconversions on aromatic rings **(16 lectures)**

Textbook

1. Carey, F. A. and Giuliano, R. M., (2012), Organic Chemistry, 8th Edition, McGraw Hill Education
2. Clayden,J., Greeves,N., and Warren,S., (2012), Organic Chemistry, 2nd Edition, Oxford
3. Sykes,P., A Guidebook to Mechanism in Organic Chemistry, 6th Ed., Longman

Reference Books

1. Smith, M. B., (2016), March's Advanced Organic Chemistry, 7th Edition. Wiley
2. Carey, F. A. and Sundberg, R. J., (2007), Advanced Organic Chemistry Part A: Structure and Mechanisms, 5th Edition. Springer
3. Carey, F. A. and Sundberg, R. J. (2007), Advanced Organic Chemistry Part B: Reactions and Synthesis, 5th Edition. Springer

Self-learning Material

1. Punniyamurthy, T., NPTEL Course material, Department of Chemistry, Indian Institute of Technology Guwahati, http://https://nptel.ac.in/courses/104103071
2. Sunoj, R. B., Organic Reaction Mechanism, NPTEL Course material, Department of Chemistry, Indian Institute of Technology Bombay, http://nptel.ac.in/courses/104101005/

Title	**Physical Organic Chemistry and Photochemistry**	Number	CYL3XXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The instructor will:

1. Provide understanding of representing the structures of organic molecules, its active sites and reactivity.
2. Provide knowledge of selective reactions, reactivity, reactions using light, photochemical methods and pericyclic reactions

Learning Outcomes

The students are expected to have the ability to:

1. Design and develop selective reactions from the knowledge of the chemical reactivity.
2. Advance their knowledge in strategizing and synthesizing of the small biologically active molecules and heterocycles using the photochemical and pericyclic pathway.
3. Appreciate chemistry of light and heat essential in numerous reactions and synthesis.

Contents

Introduction: Concepts of inductive effect, Electromeric effect, Resonance effect, Hyperconjugation, Captodative Effect (**5 lectures**)

Stereoelectronic Effects: Participation of sigma and pi bonds in transition state, Orbital symmetry and frontier orbitals, Hammond's Postulate, Hammett Plot, Curtin–Hammett principle (**8 lectures**)

Pericyclic Reactions: Introduction of pericyclic reactions, Classifications, Cycloaddition, Electrocyclic, Sigmatropic, Group transfer reactions (**15 lectures**)

Photochemistry: Basic principles of photochemistry, Photochemical reaction of cyclic and acyclic carbonyl compounds including enones and dienones, Norrish Type I and II cleavage, Photoreduction, Patterno-Buchi reaction, photochemistry of alkenes, dienes, aromatic compounds, Chemistry of vision (**12 lectures**)

Textbook

1. Clayden,J., Greeves,N., and Warren,S., (2012), Organic Chemistry, 2nd Edition, Oxford
2. Singh, J. and Singh, J., (2012), Photochemistry and Pericyclic Reactions, 3rd Edition. New Age

Reference Books

1. Smith, M. B., (2016), March's Advanced Organic Chemistry, 7th Edition. Wiley
2. Koswer, E.M., (2009), An Introduction to Physical Organic Chemistry, International Edition. Wiley
3. Carey, F. A. and Sundberg, R. J., (2007), Advanced Organic Chemistry Part A: Structure and Mechanisms, 5th Edition. Springer
4. Carey, F. A. and Sundberg, R. J. (2007), Advanced Organic Chemistry Part B: Reactions and Synthesis, 5th Edition. Springer
5. Modern Methods of Organic Synthesis, William Carruthers and Iain Coldham, Cambridge University Press

Self-learning Material

Singh, N.D. P., Organic Chemistry and Pericyclic Reactions, NPTEL Course material, Department of Chemistry, Indian Institute of Technology Kharagpur, http://www.nptel.ac.in/courses/ 104105071/

Title	**Purification and Characterization of Organic Molecules**	Number	CYL3XXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The instructor will:

1. teach basic aspects of isolation and purification techniques, spectroscopic methods, important analytical tools in academic research, chemical and pharmaceutical industries for isolation and identification of organic molecules.
2. teach elucidation of structures of organic compounds using in-depth analysis of different spectroscopic data.
3. provide a practical knowledge of organic chemistry.
4. provide an exposure to the laboratory techniques for the synthesis, purification, isolation and characterization of organic compounds
5. make students aware of laboratory safety and use of hazardous chemicals
6. importance of material safety data sheets (MSDS) which are essential for product stewardship and occupational safety and health.

Learning Outcomes

The students are expected to have the ability to:

1. determine the structures of known/unknown organic molecules utilizing several spectroscopic techniques mentioned in the content.
2. appreciate modern spectroscopic techniques and its applications in chemistry and science.
3. Learn basic experimental techniques in organic chemistry
4. Be able to carry out simple synthetic organic transformations, isolate, purification and characterization of the reaction products
5. users should be familiarizing with physical data (melting point, boiling point, flash point, etc.), toxicity, health effects, first aid, reactivity, storage, disposal, protective equipment, and spill-handling procedures.

Contents

Purification Techniques: Crystallization, steam and vacuum distillation, fractional distillation, sublimation, differential extraction, thin layer chromatography (TLC), column chromatography, HPLC **(3 lectures)**

Infrared Spectroscopy: Introduction to IR spectroscopy, Modes of stretching and bending, stretching frequencies of different organic functional groups, Effects of structural and conformational environment **(3 lectures)**

Mass Spectrometry: Principles of mass spectrometry, Different ionization techniques (EI, CI, ESI etc.), GC-MS, HRMS, LC-MS and MS-MS techniques, Fragmentation and rearrangement of different classes of organic molecules, Isotope effects, Identification of known/unknown organic compounds **(6 lectures)**

Nuclear Magnetic Resonance Spectroscopy: Chemical shifts of different organic functional groups in 1H and 13C NMR, origin and interpretation of coupling constant, introduction to the principle and interpretation of DEPT, COSY, NOESY, NOE techniques, Structural elucidation of organic compounds **(15 lectures)**

Ultraviolet-Visible Spectroscopy: Studies on conjugated and extended conjugated systems, analysis and interpretation of UV-Visible spectroscopic data, Application in mechanistic studies **(3 lectures)**

Identification and Mechanistic Studies: Application of all above techniques in the structural elucidation of unknown organic compounds and investigation of organic reaction mechanisms **(12 lectures)**

Text Books

1. Donald L. Pavia, Gary M. Lampman, George S. Kriz, James A. Vyvyan: Introduction to Spectroscopy, 4th Edition, Brookes Cole, 2008.
2. Robert M. Silverstein, Francis X. Webster, David Kiemle: Spectrometric identification of organic compounds, 7th Edition, Wiley, 2005.

Reference Books

1. Harald Gunther: NMR spectroscopy, Basic principles, concepts, and applications in chemistry, 2nd Ed., Wiley, 2001 (reprint)
2. Timothy Claridge: High Resolution NMR Techniques in Organic Chemistry, 2nd Ed. Elsevier, 2009

3.Edmond de Hoffmann, Vincent Stroobant: Mass Spectrometry, Principles and applications, 3rd Edition, Wiley, 2007

Self-Learning Material

1.http://nptel.ac.in/courses/104103071/26

Title	Organic Chemistry Lab 1	Number	CYP3XXX
Department	Chemistry	L-T-P [C]	0–0–3 [1.5]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives
The instructor will:
1. teach basic aspects of isolation and purification techniques, spectroscopic methods, important analytical tools in academic research, chemical and pharmaceutical industries for isolation and identification of organic molecules.
2. teach elucidation of structures of organic compounds using in-depth analysis of different spectroscopic data.
3. understand chemical waste disposal and segregating solvents

Learning Outcomes
The students are expected to have the ability to:
1.determine the structures of known/unknown organic molecules utilizing several spectroscopic techniques mentioned in the content.
2.appreciate modern spectroscopic techniques and its applications in chemistry and science.

Laboratory Experiments:

1.Melting point measurement of a given organic compound

2.Identification of an organic compound by HRMS

3.Identification of functional group present using IR spectroscopy

4.Identification of organic compound using NMR spectroscopy

5.Purification of an organic compounds via recrystallization

6.Separation of mixture of organic molecules by distillation techniques

7.Separation of a mixture of organic compounds by using extraction techniques

8.Chromatography techniques: TLC and Column Chromatography

9. Few crystallization techniques

10. soxhlet extraction technique

11. Solvent drying procedures

Text Books
1. Donald L. Pavia, Gary M. Lampman, George S. Kriz, James A. Vyvyan: Introduction to Spectroscopy, 4th Edition, Brookes Cole, 2008.
2.Robert M. Silverstein, Francis X. Webster, David Kiemle: Spectrometric identification of organic compounds, 7th Edition, Wiley, 2005.
3.Vogel, A. I., Textbook of Practical Organic Chemistry, Pearson

Reference Books
1.Harald Gunther: NMR spectroscopy, Basic principles, concepts, and applications in chemistry, 2nd Ed., Wiley, 2001 (reprint)
2.Timothy Claridge: High Resolution NMR Techniques in Organic Chemistry, 2nd Ed. Elsevier, 2009
3.Edmond de Hoffmann, Vincent Stroobant: Mass Spectrometry, Principles and applications, 3rd Edition, Wiley, 2007
4.Ault, A., Techniques and Experiments for Organic Chemistry, University Science Books

Self-Learning Material
1.http://nptel.ac.in/courses/104103071/26
2.https://ocw.mit.edu/courses/5-37-introduction-to-organic-synthesis-laboratory-spring-2009/pages/labs/

Title	Organic Chemistry Lab 2	Number	CYP3XXX
Department	Chemistry	L-T-P [C]	0–0–3 [1.5]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives
The instructor will:
1. provide a practical knowledge of organic chemistry.
2.provide an exposure to the laboratory techniques for the synthesis, purification, isolation and characterization of organic compounds

Learning Outcomes
The students are expected to have the ability to:
1.Learn basic experimental techniques in organic chemistry
2.Be able to carry out simple synthetic organic transformations, isolate, purification and characterization of the reaction products

Laboratory Experiments:
1. Detection of an organic functional group: aldehyde/ketone
2. Reduction reaction: Reduction of ketone to alcohol
3. Oxidation reaction: Oxidation of primary alcohol to aldehyde using PCC
4. Aromatic electrophilic substitution reaction: Bromination of aniline derivatives
5. Esterification: Synthesis of Aspirin
6. Green Reaction: Reaction in the water medium as a solvent
7. Carbon-Carbon bond formation reaction (Wittig reaction)
8. Protection of functional group (ex. alcohol/amine)

Text Books
1.Robert M. Silverstein, Francis X. Webster, David Kiemle: Spectrometric identification of organic compounds, 7th Edition, Wiley, 2005.
2.Vogel, A. I., Textbook of Practical Organic Chemistry, Pearson

Reference Books
1.Timothy Claridge: High Resolution NMR Techniques in Organic Chemistry, 2nd Ed. Elsevier, 2009
2.Ault, A., Techniques and Experiments for Organic Chemistry, University Science Books

Self-Learning Material
1.http://nptel.ac.in/courses/104103071/26
2.https://ocw.mit.edu/courses/5-37-introduction-to-organic-synthesis-laboratory-spring-2009/pages/labs/

Title	**Biochemistry**	Number	BBL2010
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	B.S. in Chemistry	Type	Compulsory
Prerequisite			

Objectives
The Instructor will:
Introduce basic chemical properties of biomolecules and principles governing chemical properties of cell.

Learning Outcomes
The students will have the ability to: Understand the structure-function relationship of biomolecules in living system

Contents

Chemistry of Lipids **[5 Lectures]**: Classes of Lipids and their structure; Membrane Lipids and their properties; Bio membrane organization;Sterols;Membrane Transport
Chemistry of Amino Acids **[5 Lectures]**: Amino acid structure and properties; Peptide bond; acid-base chemistry of amino acids; isoelectric point
Chemistry of Proteins **[5 Lectures]**: Protein structure and its importance; Enzymes; Michaelis-Menten kinetics; Enzyme inhibition
Nucleic acid chemistry **[5 Lectures]**: Nucleotides; DNA; RNA; Types of RNA
Properties of Carbohydrates **[4 Lectures]**: Aldoses, Ketoses; Epimers; Mutarotation; Glycosidic bond
Carbohydrates Chemistry **[5 Lectures]**: Common polysaccharides; Reducing & Non reducing sugars; Glycoproteins
Bioenergetics **[5 Lectures]**: Catabolism; Anabolism; Energy Coupling; High energy bonds formation and dissociation
Metabolic pathways **[5 Lectures]**: Glycolysis; TCA, Electron Transport Chain, Calvin cycle, Reverse TCA
Active learning component **[3 Lectures]**: Flipped classroom (group presentations), debates, case studies on applied biochemistry, group discussions, think-pair-share

Textbook
D. L. Nelson and M. N. Cox, Lehninger Principles of Biochemistry, WH Freeman, (2012), Sixth Edition.

Reference Books
D. Voet, C. W. Pratt and J. G. Voet, Principles of Biochemistry, Wiley (2012), Fourth Edition.

Self-Learning Material
S Dasgupta, Biochemistry I, NPTEL Course Material, Indian Institute of Technology Kharagpur, http://nptel.ac.in/courses/102105034/

Title	**Coordination and Bioinorganic Chemistry**	Number	CYL3XXX
Department	Chemistry	L-T-P [C]	3-0-0[3]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. Provide background to basic concepts and key knowledge base of Coordination and Bioinorganic Chemistry.

Learning Outcomes

The students are expected to have the ability to:

1. Understand the bonding fundamentals for both coordination compounds, predicting geometries of simple molecules.
2. Appreciate the fundamentals of bioinorganic chemistry.

Contents

Introduction: coordination geometry, coordination numbers, ligands, Werner's postulates, nomenclature, isomerism, thermodynamic stability, step-wise and overall binding constant, chelate and macrocyclic effect. **(3 Lectures)**

Theories of Bonding: VBT, CFT MOT and their limitations; splitting of d-orbitals, CFSE of d1 to d10 system. Spectral and magnetic properties; origin of colors, term symbols, selection rules for electronic transitions,Orgel diagram, calculations of Dq, B and Nephelauxetic effect, magnetochemistry **(16 Lectures)**

Inorganic reactions and mechanisms, hydrolysis reactions, substitution reactions trans-effect; isomerization reactions, redox reactions; mixed valence systems, metal-metal bonding, cluster compounds **(9 Lectures)**

Bioinorganic chemistry: Introduction, Bioinorganic chemistry of iron: hemoglobin, myoglobin, cytochromes. Chemistry of zinc, cobalt, and copper-containing metalloenzymes. **(14 Lectures)**

Textbook

1. Huheey, J.E. Keiter, E. A.; Keiter, R. L.; Medhi, O.K.; (2006), Inorganic Chemistry, Pearson; 4th edition
2. Cotton, F. A.; Wilkinson, G.; Basic Inorganic Chemistry (2007), Wiley; 3rd edition
3. S. J. Lippard (1994) Principles of Bioinorganic Chemistry, University Science Books.

Reference Books

1. Housecroft, C and Sharpe, A. G. (2012) Inorganic Chemistry, 4th Edition, Pearson

Self-Learning Material

1. Ray, D Coordination Chemistry, NPTEL course material, Department of Chemistry, Indian Institute of Technology, Kharagpur, http://nptel.ac.in/courses/104105033/
2. Maiti, D., Basics in Inorganic Chemistry, NPTEL Course Material,Department of Chemistry, Indian Institute of Technology, Bombay, https://www.youtube.com/watch?v=0ofu2inFF0k&list
3. Rath, S.P., Bioinorganic Chemistry, NPTEL Course Material,Department of Chemistry, Indian Institute of Technology,Kanpur,https://www.youtube.com/watch?v=FziKko-ZQww

Title	**Organometallics**	Number	CYL3XXX
Department	Chemistry	L-T-P [C]	3-0-0[3]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. impart comprehensive knowledge on main group, transition metal and f-block organometallic compounds and strengthen their foundation for potential application in the domain of catalysis, materials and polymers synthesis.

Learning Outcomes

The students are expected to have the ability to:

1. Understand structure bonding and reactivities of a diverse range of organometallic compounds.
2. Design and characterize organometallic compounds relevant for catalysis, materials, and polymer chemistry.

Contents

Introduction: History, various sigma/pi donor acceptor ligands, Structure and bonding in organometallic compounds, isolobal analogies, 18 electron rule **(5 Lectures)**

Main group organometallics: synthesis, structure and reactivities of s and p block organometallic compounds **(12 Lectures)**

Transition metal organometallics: Metal carbonyls, metal phosphines, olefin and acetylene complexes, alkyls and allyl complexes, metallocenes, carbenes and NHC complexes Major reaction types – oxidative addition, reductive elimination, migratory insertion, elimination isomerization and rearrangement reactions. **(16 Lectures)**

Organometallic compounds of Lanthanoids and actinoids **(2 Lectures)**

Organometallic reagents and catalysts, Medicinal applications **(7 Lectures)**

Textbook

1. C. Elschenbroich (2006), Organometallics, 3rd edition, Wiley
2. Gupta, B.D.; Elias, A. J Basic Organometallic Chemistry; Universities Press, 2nd edition

Reference Books

1. R. H. Crabtree, The Organometallic Chemistry of the Transition Metals, Wiley, 1998
2. Hartwig, J. H. Organotransition Metal Chemistry: From Bonding to Catalysis, University Science Books, 2009 (1st Ed)

Self-Learning Material

1. Samuelson, A.G., Introduction to Organometallic Chemistry, NPTEL Course Material, Department of Inorganic and Physical Chemistry, Indian Institute of Science, Bangalore, http://nptel.ac.in/courses/104108062/

2. Maiti, D., Organometallic Chemistry , NPTEL Course Material, Department of Chemistry, Indian Institute of Technology, Bombay, https://www.youtube.com/watch?v=ITfwjQemwMg&list

Title	**Chemical Kinetics, Photochemistry and Theory of Solutions**	Number	CYL3XXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. provide fundamental aspects of Laws of gasses, chemical kinetics, catalysis and photochemistry
2. discuss the application of laws for experimental determination of kinetic and photochemistry

Learning Outcomes

1. Students will understand fundamental topics in chemistry
2. Students will understand theory for experimental determination of reaction rate constant

Contents

Chemical Kinetics: Rate of reaction, Rate laws and rate constants, order of reaction - First-order, Second-order, Third-order and Zero-order, Half-life of reaction, Experimental methods to determine order, effect of temperature and catalyst on reaction rates, Arrhenius equation, Collision theory of gaseous reaction, Eyring equation, Theory of unimolecular gaseous reactions, Kinetics of complex reaction - reversible, consecutive and chain reactions, diffusion controlled reaction, influence of solvent and ionic strength on reaction, molecular reaction dynamics, potential energy surface, theoretical calculation of rate constants, The Marcus theory **(20 lectures)**

Catalysis: characteristics of catalytic reaction, enzyme catalysis, heterogeneous catalysis, effect of temperature and pH on enzyme catalysis **(4 lectures)**

Photochemistry: Jablonski Diagram, Beer-Lambert Law, Laws of photochemistry, quantum Yield, Photochemical rate laws, Kinetics of photochemical reactions, Fluorescence, energy transfer, quenching, electronic spectra and photochemistry **(10 lectures)**

Ideal solution state: solution composition, partial molar quantities, ideal solutions, thermodynamic properties of ideal solution, ideally dilute solution, thermodynamic properties of ideal dilute solution, vapor pressure, lowering of vapor pressure, elevation of boiling point, depression of freezing point, osmotic pressure, abnormal behavior of solution **(8 lectures)**

Textbook

1. Atkins, P.W.; Paula, J.de. (2014), Atkin's Physical Chemistry Ed., 10th Edition, Oxford University Press.
2. Levine, R. N. Physical Chemistry, sixth Ed., Mc Grow Hill
3. Ball, D. W. (2017), Physical Chemistry, 2nd Edition,Cengage Learning, India.
4. Castellan, G. W. (2004), Physical Chemistry, 4th Edition, Narosa.
5. D. A. McQuarrie and John D Simon,Physical Chemistry, 1st Ed, Viva Books, India (2003).

Reference Books

1. Kapoor, K.L. (2015), A Textbook of Physical Chemistry, Vol 1, 6th Edition, McGraw Hill Education.
2. Puri, Sharma and Pathania (2013), Principles of Physical Chemistry, 46th Edition, Vishal Publishing
3. R. Stephen Berry, Stuart A. Rice, John Ross, Physical Chemistry, 2nd edition, Oxford University Press

Self-Learning Material

https://nptel.ac.in/courses/112105123

Solutions (Physical Chemistry) https://youtu.be/RG7bDGfbV_Y

Title	**Thermodynamics, Chemical Equilibria, Phase Equilibria and Electrochemistry**	Number	CYL2XXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. provide fundamental aspects of Thermodynamics, chemical equilibria and phase equilibria

Learning Outcomes

The students are expected to have the ability to:

1. understand fundamental topics in chemistry which are Thermodynamics, Chemical equilibria and Phase equilibria

Contents

Thermodynamics: Work, heat & energy, First Law of Thermodynamics, Enthalpy, heat capacity, Relations between C_p and C_v, Expansion of ideal gas - Isothermal and Adiabatic, Joule-Thomson effect, Zeroth Law of Thermodynamics, changes of enthalpy and entropy in reactions, Kirchhoff equation, Hess's Law, second law of thermodynamics, spontaneous, irreversible and cyclic process, Carnot theory, entropy changes in various processes, Maxwell's relationship, Gibbs-Helmholtz equation, thermodynamics of open system, chemical potential, Third law of thermodynamics, determination of absolute entropies, entropy of real gas, residual entropy **(17 lectures)**

Chemical Equilibrium: Spontaneous reactions, Standard free energy, Chemical equilibrium, law of mass action, thermodynamics derivative of law of mass action, Van't Hoff reaction isotherm, Relation between K_P, K_C, K_X, De Donder treatment for chemical equilibria, Thermodynamic relation for chemical affinity, Homogeneous equilibria, Temperature dependence of equilibrium constant, The van't Hoff equation, Integrated form of Van'thoff equation, Pressure- dependence of equilibrium constant, Heterogeneous equilibria, Equilibrium constant for reaction involving real gases **(5 lectures)**

Phase Equilibrium: Phase**,** Components, Degree of freedom, Conditions for equilibrium between phases, The Gibbs phase rule, The Derivation of the phase rule, One-component systems: Water system, Carbon dioxide system, Sulphur system, Liquid helium system, High pressure phase diagrams: Water system and Carbon system, Two-component systems: Type A- Simple eutectic systems, Thermal analysis: cooling curves, Lead-silver system, Bismuth-cadmium system, Potassium iodide-water system, Freezing mixtures. Type B- Systems in which two components form a stable compound, Formation of compounds with congruent melting points, Formation of compounds with incongruent melting points, Calculation of eutectic point and eutectic composition, Three-component solid-liquid systems: Acetic acid-chloroform-water system, Three-component systems consisting of two salts and water, The Ehrenfest classification of phase transitions. **(10 lectures)**

Electrochemistry: Faraday's law, Types of conductance, Measurement of equivalent conductance, Variation of conductance with dilution, Kohlrausch's law, Arrhenius theory of electrolyte dissociation and its limitations.] Weak and strong electrolytes, Ostwald's dilution law, its uses and limitations – DebyeHuckel-Onsager's equation for strong electrolytes, Migration of ions and Transport number, Applications of conductivity measurements: Determination of degree of dissociation, ionic product of water and solubility product of sparingly soluble salts, Conductometric titrations, strong acid-strong base, weak acid-strong base, strong acid-weak base and weak acid-weak base. **(10 lectures)**

Textbook

1. Atkins, P.W.; Paula, J.de. (2014), Atkin's Physical Chemistry Ed., 10th Edition, Oxford University Press.
2. Levine, R. N. Physical Chemistry, sixth Ed., Mc Grow Hill
3. Ball, D. W. (2017), Physical Chemistry, 2nd Edition,Cengage Learning, India.
4. Castellan, G. W. (2004), Physical Chemistry, 4th Edition, Narosa.

Reference Books

1. Kapoor, K.L. (2015), A Textbook of Physical Chemistry, Vol 1, 6th Edition, McGraw Hill Education.
2. Puri, Sharma and Pathania (2013), Principles of Physical Chemistry, 46th Edition, Vishal Publishing
3. S. Glasstone, An Introduction to Electrochemistry, East-West Press Pvt. Ltd., New Delhi, 2007.

Self-Learning Material

Basic Thermodynamics, Dr. Suman Chakroborty, NPTEL
https://nptel.ac.in/courses/112105123
Sangaranarayanan, M.V., Electrochemistry, Department of Chemistry, IIT Madras; https://nptel.ac.in/courses/104106129

Title	**Application of Organometallic compounds in synthesis**	Number	CYL6XXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	BS in Chemistry	Type	Elective
Prerequisite			

Objectives

The Instructor will:

1. To introduce the fundamentals of organometallic compounds in Organic Synthesis with an emphasis on the modern methods.
2. Philosophy of organic synthesis, understanding of the organometallic reagents, and its role in organic synthesis and reaction mechanism.
3. Metal catalyzed method development and cascade approaches that helps in shaping academic as well as industrial research.

Learning Outcomes

The students are expected to have the ability to:

1. Understand the basic properties of organometallic compounds and their reactivity's
2. Differentiate between metals by their reactivity's and role in organic synthesis
3. Ability to design and develop the novel method/reactions utilizing organometallic catalysis

Contents

Brief history of metals B, Li, Mg, Cu, Sn, Fe, Ru, Co, Zn and Ni, biological importance, nomenclature, 18e- rule, classification, preparation and properties, general reaction – transmetallation, oxidative addition reductive elimination and migratory insertion **(6 Lectures)**

B: Hydroboration-oxidation, Rearrangements, Allylboration, as reducing agent, Suzuki reaction and related reactions. **(4 Lectures)**

Li, Cu and Mg: Grignard and Organolithium Reagents, Gillman reagent, Conjugate addition, Ullmann reaction and Applications of BuLi and MeLi in the Reactions **(5 Lectures)**

Sn: Synthesis and reactions of organotin compounds, tributyltin azide, organostannane addition, tributyltin hydride in radical reactions, stille reaction and reduction of organic compound **(5 Lectures)**

Ru: Regioselective Reductions, Oxidations, C-C bonds formation, Ruthenacycle Intermediates, Olefin Metathesis, Ring closing metathesis and Cyclopropanation**(5 Lectures)**

Fe: Polymerizations, oxidative couplings, oxidations, reductions, C- C bond formation, Ferrier rearrangement, one-pot multicomponent condensations, Friedel-Crafts reactions, Prins-type cyclisation, deprotection of various functional groups. **(5 Lectures)**

Co: Hydrogenation, hydro-functionalization, cycloaddition, radical and biomimetic reactions, cross-coupling, synthesis of heterocyclic compounds (Pauson-Khand) **(5 Lectures)**

Zn: Reductions of unsaturated compounds, oxidation, hydroamination, depolymerizations, transformations of carbon dioxide, oxidative coupling reactions, C-C, C-N, and C-O bond formation, Reformatsky reaction, Simmons–Smith reaction, Barbier reaction **(4 Lectures)**

Ni: Alkene/alkyne oligomerizations, Coupling reactions and Ni carbonylation. **(3 Lectures)**

Text Books

1. Shun-IchiMurahashi: Ruthenium in Organic Synthesis, Wiley, 2004
2. Clayden, J., Greeves, N., and Warren, S., (2012), Organic Chemistry, 2nd Edition, Oxford

Reference Books

1. L. Brandsma, S.F. Vasilevsky and H.D. Verkruijsse: Application of Transition Metal Catalysts in Organic Synthesis, 9th edition, Springer, 1999
2. Jiro Tsuji: Transition Metal Reagents and Catalysts: Innovations in Organic Synthesis, 1st Edition, Wiley, 2002

Self-Learning Material

https://nptel.ac.in/content/storage2/courses/104103022/download/module11.pdf

Title	**Theory of complex fluids**	Number	CYL4XXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. demonstrate the fundamental behaviour of complex fluids
2. explain the interactions and dynamics of non-interacting and interacting fluids

Learning Outcomes

1. Basic understanding of the complex fluids relevant in the area of chemistry, physics, material sciences and engineering
2. The concepts will be applicable to industrially relevant practical applications.

Contents

Soft matter, Comparison between hard and soft matter, liquid crystals, gels, colloids, polymer melts, surfactants, macromolecules and biological molecules, disordered and glassy systems, Relevant length and time scale, Intermolecular forces, Model interaction potentials, self-assembled macrostructures, Phase space, classical fluids, reduced configurational distribution function, Pair correlation functions, Experimental techniques for liquid structure, scattering theory and structure factors, Ornstein Zernike theory (12 lectures)

Response Functions and Fluctuations, Thermal fluctuations, Fluctuation and critical phenomena, Time-Correlation Functions, Linear Response Theory, Fluctuation-Dissipation Theorem, Diffusion, Friction, and Viscosity, Stokes-Einstein relation, Green-Kubo relations (12 lectures)

Langevin Equation, Fokker–Planck Equation, Smoluchowski Equation, Master Equations, Rouse theory, Kremer-Grest model, Supercooled fluids, Glasses, Breakdown of Stokes-Einstein relation (18 lectures)

Textbook

1. Jones, R. A. L., Soft Condensed Matter, Oxford University Press 2002.
2. Chandler, D., Introduction to Modern Statistical Mechanics, 1st Edition, Oxford University Press, 1987.
3. Robert Zwangzig, Non equilibrium Statistical Mechanics, Oxford University Press, 2001.
4. D. A. McQuarrie and John D Simon, Physical Chemistry, 1st Ed, Viva Books, India (2003).
5. B. Widom, Statistical Mechanics A Concise Introduction for Chemists, Cambridge University Press
6. Biman Bagchi, Statistical Mechanics for Chemistry and Materials Science, CRC press (2018).

Reference Book

1. Hansen, J.-P. and McDonald, I. R., Theory of Simple Liquids: With Applications to Soft Matter, 4th Edition, Elsevier, 2013
2. Biman Bagchi, Molecular relaxation in liquids, Oxford University Press, 2012
3. William Feller, An Introduction to Probability Theory and Its Applications, Volume 1, John Wiley and Sons
4. L. D. Landau and E. M. Lifshitz, Statistical Physics, 3rd edition, part1,

Self-Learning Material

1. Statistical Thermodynamics of Complex Liquids, MIT Open Course Ware Number 22.52J /8.575J / 10.44J by Prof. Sow-Hsin Chenhttps://ocw.mit.edu/courses/nuclear-engineering/22-52j-statistical-thermodynamics-of-complexliquids-spring-2004/syllabus/
2. Course Introduction Basic Statistical Mechanics by Biman Bagchi https://www.youtube.com/watch?v=M_dhvmM2fmI&list=PLOzRYVm0a65en9rkAeG_KDhh4thVEu5Od

Title	**Chemical Bonding and Statistical Rate Theories**	Number	CYL4XXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. Provide understanding of electronic Schrodinfger equation and the birth of potential energy surface
2. provide a basic idea of how to describe nuclear motion during chemical reaction

Learning Outcomes

Students are expected to:

1. have a good understanding of the concept of energy profile and reaction rates
2. understand how nuclear motion can be described using classical and quantum methods

Contents

Multielectron systems: Variational and Perturbation approaches for many-electron Schrodinger equation; Spin; Spin-Statistics Theorem; Fermions and the anti-symmetry principle; Many-electron atoms; Term-Symbols; Spin-orbit interactions **(14 lectures)**

Molecular Electronic Structure: Born-Oppenheimer approximation; Potential energy surfaces; H_2^+ ion; Valence bond (VB) and Molecular orbital (MO) approaches; Hartree-Fock SCF method; Correlation; Electron probability density; Virial Theorem; Hellmann-Feynman Theorem **(12 lectures)**

Electronic Structure: Introduction to ab initio, density-functional; semi-empirical and molecular mechanics methods **(12 lectures)**

Rate Theories: Transition state theory (TST) and Rice-Ramsperger-Kassel-Marcus (RRKM) theory **(4 lectures)**

Textbook

1. Quantum Chemistry, I. N. Levine, Pearson Education India, 7th Edition (2016).
2. R. D. Levine, (2009) Molecular Reaction Dynamics, Reprint Edition, Cambridge University Press, USA.

Reference Books

1. Modern Quantum Chemistry, A. Szabo and N. S. Ostlund, Dover publications (1996).
2. T. Baer and W. L. Hase, (1996), Unimolecular Reaction Dynamics: Theory and Experiments, Oxford University Press, USA
3. Jorge José Kohanoff, Electronic Structure Calculations for Solids and Molecules:Theory and Computational Methods, Cambridge University Press (2006)

Self-Learning Material

1. Halder, M., Reaction Dynamics, NPTEL lectures, Department of Chemistry, Indian Institute of Technology, Kharagpur, http://nptel.ac.in/courses/104105041/31
2. K. Mangala Sunder, Introduction to Chemistry & Quantum Chemical Methods, NPTEL course material, Department of Chemistry, IIT Madras, http://nptel.iitm.ac.in

Title	**Industrial Catalysis**	Number	CYL6XXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	BS in Chemistry, M.Sc.	Type	Compulsory
Prerequisite	Basics of Organometallic Chemistry and Organic Chemistry		

Objectives

The Instructor will:

1. To impart knowledge on Catalysis and its importance.
2. To teach practical applications of catalysts applied in industry.

Learning Outcomes

1. The learners are expected to gain knowledge of independent thinking in designing, characterizing and fine tuning of catalysts suitable for potential catalytic transformations.
2. Learners will be able to translate fundamental catalytic knowledge to develop technology for industrial processes.
3. Learners will have an update on current trends and challenges of catalysis towards industrial application.

Contents

Introduction: Definition and concepts of catalysis, Activity, TOF, TON, selectivity, Homogeneous vs heterogeneous catalysis **(4 lectures)**

Homogeneous catalyzed Industrial processes: oxo synthesis, carbonylation, wacker synthesis, Oligomerisation of ethylene (SHOP process), adiponitrile synthesis, polymerization, metathesis, Asymmetric catalysis, Monsanto L DOPA process, S-Naproxen, L-menthol **(10 Lectures)**

Industrial processes with Biocatalysts: Acrylamide, aspartame synthesis, pharmaceuticals, herbicides. **(4 Lectures)**

Heterogeneously catalyzed processes: Fischer Tropsch synthesis, water gas shift reaction, methanol synthesis, oxidation of propene. Refinery processes and petrochemistry: hydrotreating, catalytic cracking, hydrocracking, catalytic reforming, steam reforming, natural gas to fuel and chemicals. **(12 Lectures)**

Electrochemical reactions, fuel cell technology, methanol oxidation, Basic principles of photo catalysis, photocleavage of water, Catalytic processes with renewable materials, biofuels, biorefinery, phase transfer catalysis, catalysis reactor **(12 Lectures)**

Textbook

1. Hagen, J. Industrial Catalysis: A practical approach

Reference Books

1. Bhaduri S.; Mukesh, D. Homogeneous Catalysis: Mechanisms and Industrial Applications

Self-Learning Material

1. Industrial homogeneous Catalysis: Principles and Mechanisms by prof. debabrataMaiti, IIT Bombay, https://www.youtube.com/watch?v=GzXpURcjJwE&list=PL_uPpV3UHww2wyLk3XpR7LBqbR_2k1Fi0

Title	**Catalysis in Organic Synthesis**	Number	CYL7XXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	BS in Chemistry, M.Sc.	Type	Elective
Prerequisite			

Objectives

The instructor will:

1. Provide fundamentals of catalysis and its applications
2. Provide knowledge of different modes of catalysis, mechanistic insights and applications in organic synthesis

Learning Outcomes

The students are expected to:

1. learn the rationale behind the functioning and binding mode involved in different modes of catalysis.
2. acquire the ability to design novel catalytic systems and unfold their reactivity in different synthetic transformations.

Contents

Introduction: Fundamentals of catalysis, importance of catalysis, types of catalysis **(5 lectures)**

Biocatalysis: background of biocatalysis, Origins of enzymes, screening methods, enzyme assay, enzyme purification and characterization, Catalytic role of enzymes **(5 lectures)**

Lewis acid catalysis: main group metal salt/complexes as catalysts, transition metal salt/complexes as catalysts, precious metals as catalysts, Lanthanides as catalysts **(8 lectures)**

Metal catalysis: Cu, Pd, Pt, Rh, Ru, Ir etc in C-H activation and functionalization, coupling reactions, cycloadditions etc. **(9 lectures),**

Organocatalysis: history of organocatalysis, modes of organocatalysis, enamine, iminium, H-bonding, carbene catalysis **(9 lectures),**

Photocatalysis: principle of photocatalysis, Jablonski diagram, quantum yield, types of photocatalysts, UV-Vis photocatalysis, visible photo induced and photoredoxcatalyzed process (**6 lectures),**

Textbook

1. Gupta, BD and Elias AJ,HartwigJF,Crabtree RH Elschenbroich and Salzer
2. Hartwig, J. F. Organotransition Metal Chemistry From Bonding to Catalysis, 1st Ed.; University Science Books: Sausalito, CA, 2010.

Reference Books

1. Stanley Roberts, N J Turner, A Willetts, M Turner. (2003) Introduction to biocatalysis using enzymes and microorganisms.Cambridge university press

Self-Learning Material

Nanda, S., NPTEL Course material, Department of Chemistry, Indian Institute of Technology Kharagpur, http://https://nptel.ac.in/courses/104105032

Title	**Single Molecule Spectroscopy**	Number	CYL6XXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	BS in Chemistry, M.Sc. in Chemistry	Type	Elective
Prerequisite			

Objectives
The Instructor will:
This course will discuss advanced level theory, instrumentation and application of various single-molecule imaging techniques

Learning Outcomes:
The students are expected to:
Students will learn theory, instrumentation and lab experiments for single-molecule imaging

Contents
Introduction: History of single-molecule spectroscopy, Burst analysis, photon-counting histogram, smFRET and applications, measurements from immobilized samples, Molecular dynamics of single-molecules. Instrumentation and Microscopy **(15 lectures)**

Correlation Spectroscopy: Correlation vs cross-correlation, Theory of FCS, molecular dynamics by FCS, quantum dot and single-molecule behavior, Applications of FCS **(10 lectures)**

Applications: Confirmation Dynamics, Dynamic mechanism of biomolecule, Force and multi-parameter spectroscopy, Nanoscale-microscopy and high resolution imaging in individual cells. single-molecule enzyme Kinetics, nanopores. Application of various single-molecule imaging - tracking, ion-flux, protein-protein and protein-ligand interactions. **(12 lectures)**

Lab experiments: smFRET experiment of DNA Holliday Junction using prism-type TIRF Microscope **(5 lectures)**

Textbook
1. Grasland, Riglar and Widengren, Single-molecule Spectroscopy in chemistry, Physics and BIology, Nobel Symposium, Springer
2. Gell, Brockwell and Smith, Handbook on Single-molecule Fluorescence Spectroscopy, Oxford University Press
3. Elson and Rigler, Fluorescence correlation spectroscopy – Theory and Applications, Springer, (2012)

Reference Books
1. Lakowicz, J. R., (2006), Principles of Fluorescence Spectroscopy, 3rd Edition, Springer, New York

Self-Learning Material
iBiology, Biophysics, UCSF
https://www.ibiology.org/research-talks/biophysics/

Title	**Advanced Fluorescence Microscopy Techniques and Applications**	Number	CYL7XX
Department	Chemistry	L-T-P [C]	3–0–2 [4]
Offered for	BS, M.Sc., PhD	Type	Elective
Prerequisite			

Objectives
The Instructor will:
Light microscopy has become one of the most useful tools in the chemistry and life sciences. This course will cover basics and various methods of imaging fluorescent samples, image processing, and lab experiments.

Learning Outcomes
The students are expected to:
1. Students will learn the fundamentals of light microscopy, theory of various fluorescence imaging methods
2. Students will also learn how to do experiments with fluorescence microscopes and image analysis.

Contents
Introduction: History of light microscopy,What is light and color, Fourier space and Fourier transformation,Light as probe of matter, Properties of polarized light, Fluorescent dyes and proteins,Physics of optical components of a light microscope, Lenses and geometrical optics, Diffraction and interference in image formation, Diffraction and spatial resolution, Structured illumination, Point spread function (PSF), Numerical aperture, Resolution vs Super-resolution, Design of band-pass and dichroic filters for FRET, Bleed-through, Digital image: CCD vs EMCCD vs sCMOS vs PMT, Sensitivity vs Speed **(15 lectures)**

Microscopy: Fundamental theory and applications of Phase-contrast and Dark-field microscopy, Differential Interference contrast (DIC) microscopy, Epi-Fluorescence microscopy, Laser scanning confocal microscopy, Total Internal Reflection Fluorescence (TIRF), Two-photon microscopy, Light-sheet Microscopy, Lattice light-sheet microscopy, Super-resolution microscopy-Stochastic optical reconstruction microscopy, Video Microscopy for Single-molecule tracking and measuring dynamics, plasma membrane dynamics, DNA & RNA dynamics, single-molecule to Tissue imaging. **(20 lectures)**

Experiments: (1) DIY: Build your own microscope, (2) How to focus a sample, (3) Measurement of object size from image size, (4) Measurement of point spread-function (PSF), (5) Demonstration of Abbe Diffraction, (6) Setting up of Khelar Illumination, (7) Multi color image analysis and deconvolution by ImageJ. **(7 lectures)**

Textbook
1. Murphy, D. Fundamentals of Light Microscopy and Electronic Imaging (2001), Wiley & Sons, Inc.
2. Cornea, A and Conn, P. M. Fluorescence Microscopy (2014), Elsvier

Reference Books
1. Lakowicz, J. R., (2006), Principles of Fluorescence Spectroscopy, 3rd Edition, Springer, New York

Self-Learning Material
iBiology Microscopy Series, Ron Vale, UCSF
https://www.ibiology.org/online-biology-courses/microscopy-series/

Title	**Plasmonic Nanochemistry**	Number	CY6XX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	BS in Chemistry	Type	Elective
Prerequisite			

Objectives

The Instructor will:

1. This course provides fundamental knowledge of surface plasmons and discusses their applications in chemical synthesis.
2. The course will strive to be approachable to students from a diverse set of science and engineering backgrounds.

Learning Outcomes

The students are expected to have the ability to:

1. Understand fundamental aspects of chemistry plasmonics.
2. Understand generation and techniques to apply plasmonic technologies in chemical synthesis and environmental remediation.

Contents

Fundamental wave properties of lightIntroduction of Plasmonic: Dynamics of Plasmon Excitation and Decay, Collective Charge Dynamics, Confined Systems, Plasmonic Decay Channels, Hot-Electron Energy Distribution and Mechanisms of Generation, Charge Transfer Mechanisms Associated with Plasmons, IndirectHot Carrier Injection and Photoheating.**(8 Lectures)**

Synthesis of Plasmonic Nanoparticles: Earth Abundant Plasmonic Nanomaterials, Monometallic Plasmonic Nanoparticles, Quasispheres Nanorods, Nanocubes, Nanotriangles, Nanostars, Nanowires and Nanoplates, Superstructures **(8 Lectures)**

Characterization and Properties of Plasmonic Nanostructures: Ensemble Studies and Mechanistic Mysteries, Monitoring an Ensemble Reaction, Ensemble Experiments and Uncovering Mechanisms and Spectroscopic Techniques. **(8 Lectures)**

Applications of Plasmonics in Chemical Synthesis: Experimental Set up, Oxidation and Reduction of Nitroarenes, Hydrogenation of Alkenes and Alkynes, Hydrogenation of Aldehydes and Ketones, Photocatalytic redox Reactions and Photosynthesis. **(8 Lectures)**

Applications of Plasmonics in Environmental Remediation: Introduction, Water treatment, Air Purification, Dehalogenation, N 2 Fixation reaction, Electrocatalysis, Hydrogen Evolution Reaction, Oxygen Evolution Reaction, Oxygen Reduction Reaction and Electrocatalytic CO 2 Reduction **(10 Lectures)**

Textbook

1. Stefan A. Maier (2007), Plasmonics: Fundamentals and Applications, Springer
2. Mark Stockman, Tigran V. Shahbazyan (2014), Plasmonics: Theory and Applications, Springer

Reference Books

1. NordinFelidj, Marc Lamy de la Chapelle, (2019), Plasmonics in Chemistry and Biology, Jenny Stanford Publishing
2. Igor Tsukerman, G. Shvets (2011), Plasmonics And Plasmonic Metamaterials: Analysis And Applications, World Scientific

Self-Learning Material

http://www.digimat.in/nptel/courses/video/104101122/L57.html
https://www.youtube.com/watch?v=VjWmOVKtBNU

Title	**Atomic and Molecular Structure**	Number	CYL2XXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The instructor will:

1. Attempt to provide fundamental aspects of theory in view of foundations of quantum chemistry
2. Discuss a few applications of quantum theory to chemistry
3. Establish the relation between fundamentals of spectroscopy and quantum chemistry

Learning Outcomes

The students are expected to:

1. Understand fundamentals of quantum mechanics and its applications to chemistry
2. Solve and analyze exactly solvable problems and interpret solutions to understand and discuss symmetry, energy levels, spectroscopy, atomic orbitals and spins

Contents

The transition from classical to quantum regime, Uncertainty principle, the Schrödinger equation and its physical interpretation, Time-independent Schrödinger equation, wave functions, operators, commutator algebra and Uncertainty principle, Eigenvalues and Eigenvectors in quantum mechanics **(12 lectures)**

Particle in a one-dimensional box, Measurement outcomes and Fourier series, Postulates and theorems in quantum mechanics, Particle in a three-dimensional box- separation of variables, Wave functions and symmetry in quantum mechanics, Finite square well and tunnelling **(10 lectures)**

Classical Harmonic oscillator, The Harmonic oscillator model for a diatomic molecule and its relation to infrared spectra, Vibrations of diatomic molecules, Harmonic oscillator selection rules **(10 lectures)**

The rigid-rotator model for a diatomic molecule, Angular momentum, Rigid-rotator selection rules, The hydrogen atom solution and atomic orbitals, Introduction to electron spin- Stern-Gerlach experiment, Zeeman effect and spin-orbit coupling, Many-electron systems **(10 lectures)**

Textbooks

1. Levine, I. N., (1983), *Quantum Chemistry*, 3rd Edition, Allyn and Bacon

Reference Books

1. McQuarrie, D. A., (1983), Quantum Mechanics, 2nd Edition, University Science Books
2. Atkins, P. and Friedman (2005), R., Molecular Quantum Mechanics, 4th Edition, Oxford University Press, London
3. Pauling, L., (1985), Introduction to Quantum Mechanics with Applications to Chemistry, Dover Publications Inc.

Self Learning Material

Datta, A., Quantum Chemistry of Atoms and Molecules, NPTEL Course Material, Department of Chemistry, Indian Institute of Technology Bombay

Title	**Structural Chemistry**	Number	CYL4XXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. Provide basics of solid state materials, primarily crystalline solids and describe features and properties of certain specific materials.
2. Provide an interdisciplinary understanding of solid-state chemistry for understanding of how molecular structure affects the band structure and properties of materials.

Learning Outcomes

The students are expected to have the ability to:

1. Understand thermodynamics of solids
2. learn about crystals, symmetry, symmetry groups, defects, x-ray diffraction, microscopy.
3. Understand specific materials like binary alloys, oxides, nitrides and sulphides, perovskites, spinels.
4. Learn concepts related to band theory, band structure, electronic and optical properties.

Contents

Crystalline and amorphous solids, thermodynamics of solids, crystallization kinetics, nucleation, synthesis of materials. **[5 lectures]**

Unit Cell, Conventional unit cell, primitive unit cell, lattice and basis, Bravais lattices, lattice translation vectors. Symmetry in crystals, Group notations, Schoenflies and Hermann-Mauguin notations. Translational symmetry elements, glide plane, screw axis,relevance of symmetries. Crystal systems, point groups, space groups. **[13 lectures]**

Coordination number, defects and voids in crystals, Schottky and Frenkel defects, interstitials. Lattice planes, Miller indices, X-ray diffraction, Bragg's law, X-ray diffraction, indexing of peaks Powder X-ray diffractometers, single crystal X-ray diffractometers, X-ray crystallography, electron density maps, electron microscopy techniques.**[11 Lectures]**

Crystal structures of elements, alloys, binary compounds, nitrides, sulphides, perovskites, spinels.**[5 Lectures]**

Electronic structure of solids, origin of bands, band structure,density of states, Band structure and electrical conductivity, band gap, optical properties. **[8 Lectures]**

Textbook

1. West, A.R., (2015), Solid State Chemistry and Its Applications, 2nd edition, John Wiley & Sons
2. Lesley E Smart and Elaine E Moore, (2005), Solid State Chemistry: An Introduction, 3rd Edition, Taylor and Francis

Reference Books

1. B. D. Cullity and S. R. Stock, (2014), Elements of X-ray diffraction, 3rd edition, Pearson
2. C N R Rao and J Gopalkrishnan, (2004) New Directions in Solid State Chemistry, 2nd Edition, Cambridge.
3. L. Azaroff, Introduction to Solids (2017), McGraw Hill Education.

Self-Learning Material

Ranganathan, M, Solid State Chemistry, NPTEL Course Material, Department of Chemistry, Indian Institute of Technology Kanpur; https://nptel.ac.in/courses/104104101

Title	**Chemistry of Cultural Heritage**	Number	CYL7XX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	BS in Chemistry	Type	Elective
Prerequisite			

Objectives

The Instructor will:

1. Provide understanding and knowledge to students towards role of chemistry in conservation of heritage.
2. Appreciate fundamental methods and processes of chemistry to create technological advancement towards conservation of rich heritage.

Learning Outcomes

The students are expected to have the ability to:

1. Understand fundamental aspects of chemistry with specific reference to cultural heritage.
2. Understand chemical methods and techniques for protecting and transmitting the legacy of physical artefacts we inherited from past generations.

Contents

Introduction: Introduction to Heritage science, Materials and Artefacts, Constitutive Core Elements of Historic monuments, Fibers, Stone, Glass and Ceramics, Dyes, Pigments binding media and Varnishes (8 Lectures)

Identification Techniques: Infrared and Raman Spectroscopy, FORS, Energy dispersive X-ray fluorescence analysis, Atomic Absorption Spectroscopy, Thermal Analysis Techniques, Mass spectrometry, NMR and Imaging Techniques. (14 Lectures)

Separation Techniques: High Performance Liquid Chromatography and Gas Chromatography and Capillary Electrophoresis, (4 Lectures)

Case Studies (16 lectures)
Case I: Unravel Building Stone Decay Mechanisms in Urban Environments
Case II: Geochemistry in Provenance Determination of the White Marbles Used in the Ancient Great Theatre
Case III : Case Studies: Investigation of Mortars by Infrared and Raman Spectroscopy.
Case IV: Spectroscopic Measurements for the Corrosion Behaviour Evaluation of Epoxy.
Case V: Understanding and Restoration of Old Painting of Rajasthan.

Textbook

1. Evangelia A. Varella (2013), Conservation Science for the Cultural Heritage, Springer
2. Rocco Mazzeo (2017), Analytical Chemistry for Cultural Heritage, Springer
3. Chemical Analysis in Cultural Heritage, Luigia Sabbatini and Inez Dorothé van der Werf (eds.), De Gruyter Publishers (Berlin), 2020
4. Michael J. Malin (2021), The Chemistry and Mechanism of Art Materials: Unsuspected Properties and Outcomes, CRC Press.

Reference Books

1. Klaas Jan van den Berg Aviva Burnstock Koen Janssens Robert van Langh Jennifer Mass Austin Nevin Bertrand Lavedrine Bronwyn Ormsby Matija Strlic (2021) Cultural Heritage Science Springer Book Series, Springer

Self-Learning Material

1. https://nptel.ac.in/courses/124105003
2. https://www.youtube.com/watch?v=ekNo4vZwkLA

Title	**Theoretical Account of Spectroscopy**	Number	CYL4XXX
Department	Chemistry	L-T-P [C]	3–0–0
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. Introduce to the basic principles of physical chemistry, quantum mechanics and classical electrodynamics to understand molecular spectroscopy.
2. Enable students to apply quantum mechanics to understand light-matter interaction as well as application of various spectroscopic techniques.

Learning Outcomes

The students are expected to have the ability to:

1. Understand the theoretical background of various spectroscopic techniques.
2. Connect quantum mechanical description of atomic and molecular structure and spectra of atoms and molecules over a wide range of wavelength.

Contents

Time dependent quantum mechanics: Electromagnetic waves and polarization, Macroscopic view of light-matter interaction, Quantum description of matter, Time dependent perturbation theory, Fermi's Golden rule, Hydrogen atom, many-electron atoms, atomic term symbols, selection rules **(8 lectures)**

Molecular Symmetry – Symmetry operations, point groups, character table, spectroscopic term symbols for describing atomic and molecular states, total representation of a group, symmetry and dipole moment **(4 lectures)**

Theory of Rotational Motion: Classical description of molecular rotation, Rotational levels and transition rules, Rotational energy levels of diatomic molecules, energy levels of spherical and symmetric tops, moment of inertia, calculations of bond lengths and bond angles of simple triatomic molecules, vibrational stretching and vibrational satellites, no-rigid rotor, centrifugal distortion, degeneracies and intensities, Stark effect, selection rules, rotational spectra of polyatomic molecules **(10 lectures)**

Theory of Molecular Vibrations: classical description of molecular vibrations, the classical harmonic oscillator, quantum mechanics of molecular vibration, vibrational selection rules, anharmonic vibrations and Morse oscillator, bond dissociation energies and Birge-Sponer plots, calculation of force constants from vibrational spectrum, isotopic shift, rotational structure in vibrational spectra of diatomic molecules, vibration of polyatomic molecules, normal modes, Rotation-vibration spectra of diatomic molecules and calculation of molecular parameters **(10 lectures)**

Theory of Electronic Transition: Born-Oppenheimer approximation, energy of electronic transition, selection rules, the Franck-Condon principle, Electronic transitions of polyatomic molecules, Molecular orbitals, Rovibronic transitions, vibrational coarse structure, Russel-Saunders spin-orbit coupling, absorption intensity, probability of light absorption, electronic spectra, classification of electronic transition, d-d and CT transitions **(10 Lectures)**

Textbook

1. Bernath, P. F., (2005), *Spectra of Atoms and Molecules*, 2nd Edition, Oxford University Press
2. Hollas, J.M (2013), Modern Spectroscopy, Wiley

Reference Books

1. Bunker, P.R., and Jensen, P. 2012. Molecular Symmetry and Spectroscopy. E-book edition. NRC Research Press, Ottawa, Ontario, Canada. 748 p Hollas, J. M. 2013, High Resolution Spectroscopy, Elsevier Science
2. P. Atkins, R. Friedman, Molecular Quantum Mechanics (5th Ed), Oxford University Press (2010)

Self-learning Material

1. Mangal Sundar, K., *Introduction to Quantum Chemistry and Molecular Spectroscopy,* NPTEL Course Material, Department of Chemistry, Indian Institute of Technology Madras, http:// http://nptel.ac.in/courses/104106083
2. Dutta, A., *Molecular Spectroscopy: A Physical Chemist's perspective*, NPTEL Course Material, Department of Chemistry, Indian Institute of Technology Bombay, http:// https://nptel.ac.in/courses/104101099

Title	**Fundamentals of Molecular Spectroscopy**	Number	CYL3XXX
Department	Chemistry	L-T-P [C]	3–0–2 [4]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. Introduce to a few spectroscopic methods used as analytical techniques in various fields of science and engineering.
2. Introduce to theoretical background of these methods along with their instrumentation and application.
3. Enable students to use such methods whenever needed by giving hands on experience with available instruments in the institute.

Learning Outcomes

The students are expected to have the ability to:

1. Understand the theoretical background of various spectroscopic techniques.
2. Apply spectroscopic methods as analytical tool in chemistry, biology, material science, medicine.
3. Operate various spectrometers available in the institute.

Contents

1. Introduction to Spectroscopy – properties of electromagnetic radiation – interaction of radiation with matter, Einstein A and B coefficient, Beer's Law–intensity of spectral line and line shape - Line broadening, Continuous wave and Fourier Transform spectroscopy, Selection rule for photon-molecule interaction – symmetry conservation **(10 Lectures)**
2. Electronic Spectroscopy –Basic principle – Born-Oppenheimer approximation, vibrational coarse structure, Frank-Condon principle, Jablonski diagram, various photophysical processes, quantum yield expression, Chromophores, types of electronic transition, effect of conjugation, solvent. **(8 Lectures)**
3. Rotational Spectroscopy - Introduction, Rigid rotor model, Diatomic rotations, Introduction to polyatomic rotational spectroscopy **(4 Lectures)**
4. Vibrational Spectroscopy – Basic principle – Harmonic Oscillator, Morse potential, vibrational modes of diatomic and polyatomic molecules, Normal modes of vibration of CO2 molecule, Rotation-vibration spectra of diatomic molecules and calculation of molecular parameters, IR and Raman spectra of molecules, Scattering of light, Stokes and anti-Stokes shift **(8 lectures)**
5. Magnetic Resonance Spectroscopy – Origin of spin and magnetic moment, Zeeman splitting and gyromagnetic ratio, Electron and nuclear spin resonance, induction of magnetization and Gouy balance, NMR chemical shift and spin-spin coupling, introduction to AB, AX, AMX spin system, 1D 1H and 13 NMR experiments, Lanthanide shift reagents, chemical exchange. **(12 Lectures)**

Laboratory Experiments:

1.Measurement of steady state fluorescence anisotropy to determine binding constant of protein-ligand interaction.
2.Evaluation of ligand-protein interaction by fluorescence quenching analysis.
3.Study of protein secondary structures and folding kinetics by Circular Dichroism.
4.NMR chemical shift titration using 2D HSQC experiment to detect ligand-protein interaction
5.Analysis of caffeine content in cold drinks using IR spectroscopy
6.UV analysis of enzyme inhibition: Trypsin inhibition by BPTI

Textbook

1. Gunther, H., (1995), *NMR Spectroscopy*, 2nd Edition, Wiley
2. Banwell, C. N., (1994), *Fundamentals of Molecular Spectroscopy*, 3rd Edition, Tata McGraw-Hill Book Company
3. Lakowicz, J. R., (2006), *Principles of Fluorescence Spectroscopy*, 3rd Edition, Springer, New York

Reference Books

1. Bernath, P. F., (2005), *Spectra of Atoms and Molecules*, 2nd Edition, Oxford University Press
2. Silverstein, R. M. and Webster, F. M., (2010), *Spectroscopic Identification of Organic Compounds*, 6th Edition, Wiley

Self-learning Material

1. Mangal Sundar, K., *Introduction to Quantum Chemistry and Molecular Spectroscopy,* NPTEL Course Material, Department of Chemistry, Indian Institute of Technology Madras, http:// http://nptel.ac.in/courses/104106083

2. Trivedi, V. and Chaudhary, N., Bioanalytical Techniques and Bioinformatics, Department of Biotechnology, Indian Institute of Technology Guwahati, http:// http://nptel.ac.in/courses/ 102103044.

Title	**Fundamental Inorganic chemistry**	Number	CYXXXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	B.S.	Type	Elective
Prerequisite			

Objectives
The Instructor will:
1. Provide background to basic concepts and key knowledge base of general chemistry and inorganic chemistry to prepare students for careers as professionals in the field of chemistry

Learning Outcomes
The students are expected to have the ability to:
1.the bonding fundamentals for both ionic and covalent compounds, including electronegativities, predicting geometries of simple molecules.
2.the fundamentals of the chemistry of the acid-base, and important real world applications.
3.the basic knowledge about redox processes and nuclear chemistry and its applications.

Contents
Multi-electron atoms – periodic trends; chemical bonding; ionic solids; (8 Lectures) Lewis model (2 Dim); VSEPR (3 Dim) hybridization, VBT, molecular orbital theory: heteronuclear diatomics, triatomics; shapes of main group compounds **(10 Lectures)**

Acid-base chemistry: concepts, measures of acid-base strength, salts, pH, buffers, electrolytes Ionization and Hard Soft Acid and Base, acid base chemistry in non-aqueous solvent, ionic liquid, supercritical fluid **(7 Lectures)**

Oxidation and Reduction: oxidation number, reduction potential, half cell, electrochemical series, Nernst equation, disproportionation reaction, effect of complex formation, precipitation on reduction potential, chemical extraction **(7 Lectures)**

Introduction of nuclear chemistry: Define and describe the following terms, radioactivity, nuclear reactions, half life, atomic fission, atomic fusion, units of measurement of radiation **(10 Lectures)**

Textbook
1. John E. McMurry, Robert C. Fay (2015), Chemistry, Pearson
2. Lee, J.D. (2012), Concise Inorganic Chemistry, Oxford University Press; 6th edition

Reference Books
1.Inorganic Chemistry: Principles of Structure and Reactivity, 4th Edition, Pearson

Self Learning Material
1.https://nptel.ac.in/courses/104/101/104101121/
2. https://www.youtube.com/watch?v=OUj4j6td1es

Title	**Sustainable Catalysis**	Number	CYL7XXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	BS in Chemistry, M.Sc.	Type	Compulsory
Prerequisite	Basics of Organometallic Chemistry and Organic Chemistry		

Objectives

The Instructor will:

1. transmit knowledge on catalysis relevant towards a sustainable society and energy crisis.
2. demonstrate the contribution of catalysis towards circular chemistry enabling a circular economy.

Learning Outcomes:

3. The learners are expected to gain knowledge of independent thinking in designing, characterizing and fine tuning of catalysts suitable for potential sustainable catalytic transformations.

4. Learners will have an update on current trends and challenges of catalysis towards waste free product formation from readily available renewable starting materials, value-added products from waste and alternative renewable energy.

Contents

Introduction: Definition and principles of green chemistry, circular chemistry, sustainability, role of catalysis towards sustainable developments, atom economy, renewable and non-renewable materials, waste minimization, green solvents, types of catalysts, efficacy of catalysts, recycling of catalysts **(4 lectures)**

Towards sustainable chemical synthesis: transition metal catalysed hydrogenation, dehydrogenation, dehydrogenative coupling, cross coupling reactions, hydrofunctionalization of olefins, catalytic C-H functionalization, metathesis reactions, Organocatalysts in sustainable synthesis, biomimetic catalysis for green solutions, Heterogeneous catalysis in sustainable synthesis, photoredox and electrocatalytic processes (**18 lectures**)

Utilization of renewable raw materials and waste: Utilization of CO_2, N2O, depolymerisation of plastics, chemicals and fuels from biomass, lignocellulose, glycerol, sustainable synthesis of NH_3, water oxidation, hydration reactions, NO_x removal (**10 Lectures**)

Sustainable Energy: energy resources and utilization, energy density, fossil fuel, natural gas, drawbacks, renewable energy sources, energy storage, hydrogen economy, methanol economy, liquid organic hydrogen carrier, waste to energy conversion, water splitting, biomass, biofuel (**10 Lectures**)

Textbook

1. Catalysis for sustainability, Goals, challenges and impacts by Thomas P Umile, CRC Press
2. Homogeneous Catalysis with Renewables by Arno Behr and Andreas J Vorholt; Springer (2017)

Reference Books

1. Pierluigi Barbaro, Claudio Bianchini, Catalysis for Sustainable Energy Production, Wiley
2. Heterogeneous Catalysis in Sustainable Synthesis by Anne Kokel Elsevier
3. Sustainable Catalysis: Energy-Efficient Reactions and Applications by Rafael Luque

Self-Learning Material

1. Sustainable Chemistry through Catalysis, GIAN Course material, Indian Institute of Technology Kanpur, https://www.youtube.com/watch?v=TLRgBSAoJ0Y&list=PL48UwQJyfW3RreYciLRweZPZb8aygFLog&index=1
2. Xile Hu, Angamuthu, R and Bera, J. K., Catalysis for Energy Storage, GIAN Course material, Indian Institute of Technology Kanpur, https://www.youtube.com/watch?v=LvTfsIAI-VE

Title	**Entrepreneurship for Chemists**	Number	CYL3XXX
Department	Chemistry	L-T-P [C]	2–0–0 [2]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. provide an idea of entrepreneurship and how is applicable for a chemistry
2. provide a few successful case studies

Learning Outcomes:

Students are expected to

1. have an idea of what is entrepreneurship and how it can be effectively applied by a chemist
2. understand tools and techniques that prepare them for innovation and inventions.

Contents

Innovation and Invention: The need for innovation: why to innovate, what to innovate, how to innovate, and who innovates; three levels of innovation – individual, team, organization; desired qualities of individuals innovators; build innovative teams and the innovative organization dynamics Pasteur's Quadrant, Learning curves, Overview of patents, copyrights, trade secrets, and trademarks, and other forms of IP such as Geographical Indication, Elements and limitations of patentability; process of applying for a patent, Introduction to Patent search and IP diagnostic **(8 lectures)**

Concepts in Entrepreneurial Practice: Introduction to Entrepreneurship: Opportunity Identification – Generating ideas, entrepreneurial orientation and personal traits, and theories of Entrepreneurship, Starting Your Own Business: Developing an Effective Business Model, Business Model Canvas and Value Proposition Canvas, Internal & External Analysis: Industry and Competitor Analysis, Building a New-Venture Team: Equity sharing, sweat equity, founder's agreement, matching skills of team members, Market sizing, Components of a business plan, Prototype, and Minimum Viable Product **(8 lectures)**

Case Studies: **(12 lectures)**

1. Oxo process: Fischer Tropsch process, Utility of cobalt catalysts. $HCo(CO)_4$
2. Noyori asymmetric hydrogenation
3. Olefin metathesis
4. 3,4-dihydroxyphenylalanine (DOPA)
5. Aspirin

Textbook

1. Chemistry Entrepreneurship, Javier Garcia-Martinez, Li Kunhao, Wiley-VCH (2021)
2. Maital, S., & Seshadri, D. V. R., 2012. Innovation management: strategies, concepts and tools for growth and profit. SAGE Publications India.
3. Stokes, Donald E. (1997). Pasteur's Quadrant – Basic Science and Technological Innovation. Brookings Institution Press. p. 196. ISBN 9780815781776. (Pasteur's Quadrant – Basic Science and Technological Innovation.)

Reference Books

1. Technology Entrepreneurship: Taking Innovation to the Market Place, Thomas N. Duening, Academic Press, 2020
2. https://patentscope.wipo.int/search/en/search.jsf
3. https://www.wipo.int/ipdiagnostics/en/index.html
4. https.wwww.wipo.int/ipdiagnostics-assessment/
5. Academic Entrepreneurship, Michele Marcolongo, Wiley (2017)

Self-Learning Material

1. https://www.youtube.com/watch?v=vX6oYzSLw0s
2. https://www.youtube.com/watch?v=LqG-QzB-RUA

Title	**Chemistry of Elements**	Number	CYXXXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	B.S.	Type	Elective
Prerequisite			

Objectives
The Instructor will:
1. provide knowledge on the main group, transition metal and f block element based important compounds, and their application in practical purposes.

Learning Outcomes
The students are expected to have the ability to:
1. Understand fundamental aspects of main group, transition metal, lanthanide, actinide, compounds.
2. Appreciate the potential of periodic elements in practical application.

Contents

Introduction molecular geometry, symmetry and point groups, hypervalency **(8 Lectures)**

Main group compounds : Hydrogen and its compounds -ionic, covalent, and metallic hydrides, hydrogen bonding; chemistry of lithium, beryllium, boron, carbon, nitrogen, oxygen and halogen groups; chains, rings, and cage compounds. **(14 Lectures)**

Compounds of d block elements: descriptive chemistry of transition metals, periodic trends, compounds of various oxidation states of transition metals **(8 Lectures)**

Chemistry of lanthanides and actinide elements, shift reagents, strong magnets, fluorescence,. **(4 Lectures)**

Introduction to various characterization techniques for inorganic compounds: IR, multinuclear NMR, EPR, Mass. **(8 Lectures)**

Textbook
1.Cotton, F.A.; Wilkinson, G.; (2007) Advanced Inorganic Chemistry, Wiley, 6th edition
2. Lee, J.D. (2012), Concise Inorganic Chemistry, Oxford University Press; 6th edition
3. Chemistry of the Elements" Elsevier Science, Author: N. N. Greenwood, A. Earnshaw

Reference Books
1.Housecroft, C and Sharpe, A. G. (2012) Inorganic Chemistry, 4th Edition, Pearson
2. The Chemistry of the p-Block Elements:' by Prof. A. J. Elias

Self Learning Material

Balakrishna, M. S., Chemistry of Main Group Elements, NPTEL Course Material, Department of Chemistry, Indian Institute of Technology Bombay, http://nptel.ac.in/courses/117101105/

Title	**Inorganic Chemistry Lab 1**	Number	CYPXXX
Department	Chemistry	L-T-P [C]	0–0–3 [1.5]
Offered for	BS Program	Type	Compulsory
Prerequisite			

Objectives
The Instructor will:
1. impart knowledge on the different techniques of reaction set-up involved in the synthesis, isolation and purification of main group and transition metal inorganic compounds.
2. Expose students to various spectroscopic identification and characterization techniques.

Learning Outcomes
The students are expected to have the ability to:
1. Understand the basic principles behind the synthesis and properties of inorganic compounds.
2. Plan and Conduct experiments for synthesis and characterizing inorganic compounds.

Contents
Laboratory Experiments:
1. Synthesis and characterization of bispyridine iodide nitrate
2. The preparation of pentammineaquocobalt(III) chloride
3. Soliod phase Synthesis and characterization of trans bis glycinato Cu(II)
4. The preparation of two crystalline binary transition metal p-toluenesulfonate salts of [MII(OH2)6]-[OTs]2 stoichiometry, Where, MII = Co (1) and Ni (2).
5. Syntehsis and characterization of NiCl2(PPh3)2 complex
6. Color effects due to ligand-exchange in nickel complexes (Influence of ligand field tetragonality on the ground state spin of nickel (II) complexes with NH3 and H2O and ethylene diamine.)
7. Preparation of hexamminenickel(II) chloride: estimation of ammonia and nickel by titrimetric and gravimetric methods
8. The preparation of cis-and trans -potassiumdioxalatodiaquochromate(III)
9. Synthesis and characterization of acetylferrocene from ferrocene
10. Acidic and basic salts: Hydrolysis of salts and its consequences

Textbook

1. G. S. Girolami, T. B. Rauchfuss, R. J. Angelici Synthesis and Technique in Inorganic Chemistry, University Science Books, 3rd Ed, 1999.
2. Elias, A. J., A Collection of Interesting General Chemistry Experiments, Universities Press(India) Pvt. Ltd., 2002

Reference Books

1. J. Derek Woolins, Inorganic Experiments, 3rd Edition, Wiley, 2010.
2. Roesky, H. W.; Muckel, K., Chemical Curiosities: spectacular experiments and inspired quotes, VCH, 1996.
3. Handouts prepared for the laboratory experiments: collections from various literature sources (ACS Chemical Education)

Self Learning Material
John Dolhun. 5.301 Chemistry Laboratory Techniques. January IAP 2012. Massachusetts Institute of Technology: MIT Open Courseware, https://ocw.mit.edu.

Title	**Inorganic Chemistry Lab 2**	Number	CYPXXX
Department	Chemistry	L-T-P [C]	0–0–3 [1.5]
Offered for	BS Program	Type	Compulsory
Prerequisite			

Objectives
The Instructor will:
1. impart knowledge on the different techniques of reaction set-up involved in the synthesis, isolation and purification of inorganic compounds.
2. Expose students to various spectroscopic identification and characterization techniques.

Learning Outcomes
The students are expected to have the ability to:
1. Understand the basic principles behind the synthesis and properties of inorganic compounds.
2. Plan and Conduct experiments for synthesis and characterizing inorganic compounds.

Contents
Laboratory Experiments:
1. Color effects in aqueous systems containing divalent 3d metal ions
2. Preparation of nitro- and nitrito-pentamminecobalt(III) chloride
3. Aquation of trans-dichloro-bis(1,2-diamminoethane)cobalt(III)chloride
4. The Preparation and resolution of tris(ethylenediamine)cobalt(III)ion into its optical antipodes
5. Synthesis and reactions of organocobaloximes
6. Synthesis and characterization of RuHCl(CO)(PPh3)3 Introduction to organometallic synthesis under inert atmosphere.
7. Preparation of TiO2 from Titanium Isopropoxide
8. Preparation and Characterization of Hydroxyapatite
9. Preparation and magnetic properties of Fe(III)(acac) complex
10. FLP H_2 activation and hydrogenation reactions

Textbook
1. G. S. Girolami, T. B. Rauchfuss, R. J. Angelici Synthesis and Technique in Inorganic Chemistry, University Science Books, 3rd Ed, 1999.
2. Elias, A. J., A Collection of Interesting General Chemistry Experiments, Universities Press(India) Pvt. Ltd., 2002

Reference Books
1. J. Derek Woolins, Inorganic Experiments, 3rd Edition, Wiley, 2010.
2. Roesky, H. W.; Muckel, K., Chemical Curiosities: spectacular experiments and inspired quotes, VCH, 1996.
3. Handouts prepared for the laboratory experiments: collections from various literature sources (ACS Chemical Education)

Self Learning Material
John Dolhun. 5.301 Chemistry Laboratory Techniques. January IAP 2012. Massachusetts Institute of Technology: MIT Open Courseware, https://ocw.mit.edu.

Title	**Physical Chemistry Lab 1**	Number	CYPXXX
Department	Chemistry	L-T-P [C]	0–0–3 [1.5]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor(s) will:

1. Connect the basic principles of physical chemistry with experiments.
2. Enable students to apply theoretical understandings of various concepts to analytical techniques.

Learning Outcomes

The students are expected to have the ability to:

1. Design, execute and analyse experiments and their outcome.
2. Operate various analytical instruments available in the Institute.

Contents

Topics to be covered in the laboratory experiments:

- pHmetry:

Determination of pK_a values of weak monobasic, dibasic, and polybasic acid by pH-metric method (e.g. using acetic acid, succinic acid, oxalic acid and phosphoric acid etc.). ii) Determination of phosphoric acid content in soft drinks: Use of pH sensor and titrimetric analysis

- Kinetics

Determination of rate constant for first order acid catalysed hydrolysis of an ester (V_0 and $V\propto$ be supplied). ii) Determination of rate constant of decomposition of H_2O_2 by acidified KI solution using a clock reaction. iii) Determination of equilibrium constant of the reaction $KI + I_2$ KI_3 by partition method (partition coefficient to be supplied).

- Electrochemistry

Determination of E_0 of Fe^{+3}/Fe^{+2} couples in hydrogen scale by potentiometric titration of ferrous ammonium sulfate solution using $KMnO_4$ or $K_2Cr_2O_7$ as standard. ii) Study of a reversible redox reaction using Cyclic Voltammetric analysis

- Titration

Estimation of sodium carbonate using standardized HCl. ii) Estimation of carbonate and hydroxide present together in a mixture. (iii) Estimation of carbonate and bicarbonate present together in a mixture. (iv) Estimation of free alkali present in different soaps/detergents

- Solid State:
  Indexing of a given powder diffraction pattern of a cubic crystalline system.

Textbook

1. Khosla, B.D.; Garg, V.C.; Gulati, A. (2015),Senior Practical Physical Chemistry, R. Chand & Co, New Delhi
2. Kapoor, K.L. (2019),A Textbook of Physical Chemistry, Vol.7, 1st Edition, McGraw Hill Education.
3. Garland, C. W.; Nibler, J. W.; Shoemaker, D. P.(2003),Experiments in Physical Chemistry, 8th Edition, McGraw-Hill, New York.

Reference Books

Ghosal, Mahapatra and Nad (2014) – An Advanced Course in Practical Chemistry, New Central Book Agency.

Title	**Physical Chemistry Lab 2**	Number	CYPXXX
Department	Chemistry	L-T-P [C]	0–0–3 [1.5]
Offered for	BS in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor(s) will:

1. Connect the basic principles of physical chemistry with experiments.
2. Enable students to apply theoretical understandings of various concepts to analytical techniques.

Learning Outcomes

The students are expected to have the ability to:

1. Design, execute and analyse experiments and their outcome.
2. Operate various analytical instruments available in the Institute.

Contents

Topics to be covered in the laboratory experiments:

Spectroscopy: UV-Vis, Fluorescence, NMR, IR, TCSPC

- Determination of stoichiometry of complexes that are formed in solution due to incomplete equilibrium processes: Use of spectrophotometer and using Job's method of continuous variation

- Solvent effect on 1H NMR spectrum: How solvent can change the chemical shift of 1H NMR spectrum (Benchtop NMR) iii) Determination of solvent effect and fluorescence lifetime by TCSPC

Calorimetry: DSC polymers and metals

- To determine the heat capacity, glass transition temperature, and the change in heat capacity for glass transition temperature for polymers (polystyrene) by using Differential Scanning calorimetry (DSC) (ii) Determination of Heat Capacity of Metals and Demonstration of Law of Dulong and Petit using Differential Scanning Calorimeter

Phase equilibrium

- To study the **phase diagram** of a binary system (Phenol/ water) and the effect of impurities (e.g. NaCl) ii) Determination of **ionization constant** of a weak acid by conductometric method.

Surface tension

- Determine the surface tension of aqueous solutions by (i) drop number (ii) drop weight method. ii) Study the variation of surface tension with different concentrations of detergent solutions. Determine CMC.

Viscosity measurement

- Determination of **viscosity coefficient** of a given solution with Ostwald's viscometer considering aqueous solution of cane sugar, glycerol, ethanol etc.
- Study the variation of coefficient of viscosity with different concentrations of Poly Vinyl Alcohol (PVA) and determine molar mass of PVA.

Textbook

1. Khosla, B.D.; Garg, V.C.; Gulati, A. (2015), Senior Practical Physical Chemistry, R. Chand & Co, New Delhi
2. Kapoor, K.L. (2019), A Textbook of Physical Chemistry, Vol.7, 1st Edition, McGraw Hill Education.
3. Garland, C. W.; Nibler, J. W.; Shoemaker, D. P.(2003), Experiments in Physical Chemistry, 8th Edition, McGraw-Hill, New York.

Reference Books/Journal

Journal of Chemical Education

Ghosal, Mahapatra and Nad (2014)–An Advanced Course in Practical Chemistry, New Central Book Agency.